Treasury Supply News, Yield-Curve Transmission, and Macro-Financial Effects

Dhruv Kohli

Abstract

This paper studies how Treasury markets and macro-financial variables respond to exactly timed Treasury supply-news announcements. Using Treasury Quarterly Refunding documents and coupon-auction announcements from 2002–2026, I build a source-audited event database and measure high-frequency reactions with WRDS Treasury ETF minute data. The baseline proxy is constructed at the information-release level, so multiple securities announced at the same timestamp do not duplicate the same market reaction. A one-unit expanding-standardized market-implied supply-news proxy raises same-month Treasury yield changes by about 3.2–4.1 basis points; later monthly yield-change coefficients turn negative for some maturities. The current-vintage ACM decomposition is mixed, while credit and real-activity responses are less precise. The proxy is informative but noisy: calendar-matched non-event reactions remain large relative to true release-window reactions. I therefore interpret the estimates as reduced-form responses to market-implied Treasury supply news. Expected-size surprises, Databento futures, and state-dependent projections provide supporting interpretation rather than the baseline identifying variation.

1 Introduction

Treasury issuance is not only a fiscal financing operation. It changes the amount and maturity composition of safe duration risk that private balance sheets must absorb. When the Treasury announces refunding plans or coupon-auction sizes, prices can therefore move even if monetary policy and macroeconomic news are unchanged. The empirical challenge is that these announcements arrive in active Treasury markets, where ordinary intraday rate volatility can be large. This paper asks how yields, term premia, credit conditions, and real activity respond to a source-audited, high-frequency Treasury supply-news proxy while keeping that measurement problem visible.

The empirical design centers on timestamped Treasury information releases. The event set covers Quarterly Refunding Announcement (QRA) policy statements and coupon-auction announcements from 2002–2026. I collapse all securities announced at a common timestamp into a single release group, so multi-security coupon announcements contribute one market reaction rather than repeated copies of the same ETF return. WRDS TAQ Treasury ETF minute data provide the baseline market reaction, while Databento Treasury futures provide a separate 2011–2026 robustness panel. Primary-dealer expected-size surveys and TBAC recommended-financing tables are merged only after source and filtration checks, so expected-size surprises can help interpret the proxy without replacing it.

The main empirical finding is a positive same-month Treasury-market response. A one-unit expanding-standardized WRDS ETF supply-news proxy raises the 2-year, 5-year, 10-year, and 30-year monthly yield changes by about 3.2–4.1 basis points in the shock month. Later monthly yield changes turn negative for several maturities, consistent with partial subsequent reversal in yield levels but not by itself proving a cumulative level reversal. The current-vintage ACM decomposition is mixed: the same-month risk-neutral component is sharper than the term-premium component, and the paper therefore does not claim a clean term-premium channel. Credit and real-activity responses are directionally plausible but less precise.

The identification evidence is deliberately conservative. True Treasury release groups have larger average absolute reactions than matched non-event windows, but calendar-matched non-event reactions are still 83.2 percent as large as true release reactions. The current volatility-regime matched extension is appendix-gated because the reusable WRDS pseudo-event windows have insufficient intrawindow trade coverage under the finalized sample. The paper therefore estimates responses to a noisy but disciplined market-implied Treasury supply-news proxy, not to an observed structural innovation.

The methods are layered. Event timing defines the shock proxy; lag-augmented local projections estimate dynamic responses; expected-size first stages and Anderson–Rubin intervals determine whether LP-IV estimates are interpretable. Factor controls, matched non-events, narrow windows, Databento futures, and state-dependent projections assess robustness and mechanisms. State-space, spectral, option-IV, CTD, exact-holdings, and structured-expectation extensions remain outside the reported evidence unless they add reproducible information beyond the baseline.

The baseline object is the projection response to the standardized proxy

$$\theta_{i,h} = \frac{\partial \mathbb{E}[y_{i,t+h} \mid z_t, W_{t-1}]}{\partial z_t}, \qquad h = 0, \dots, H,$$

where z_t is the market-implied Treasury supply-news proxy, W_{t-1} is the predetermined control set, and $y_{i,t+h}$ is an outcome such as a yield change, term-premium change, credit spread, investment growth, or industrial production growth. When y is a monthly first difference, the reported coefficient is a response in that monthly change, not a cumulative response in the underlying level. When an observed supply quantity x_t can be credibly defined, I estimate secondary LP-IV responses to x_t. The main result remains the direct LP response to z_t, with the normalization stated explicitly.

The paper proceeds as follows. Section 2 places the design in the Treasury-supply and high-frequency identification literatures. Section 3 describes the event database and market-reaction data. Section 4 constructs the proxy. Sections 5 and 6 state the identifying assumptions and LP design. Section 7 reports the validation evidence, and Section 8 presents the baseline responses, LP-IV placement, and selected mechanisms.

2 Related Literature

The closest paper is Bi et al. (2026), who use high-frequency Treasury futures price changes around auction announcements to identify debt-volume and maturity-adjustment shocks. Their decomposition motivates this project's debt-volume and maturity-composition interpretation, but the present paper states a more conservative main claim: the baseline object is a market-implied Treasury supply-news proxy. Relative to that benchmark, this paper adds a longer WRDS ETF baseline sample, a source-audited QRA/coupon event database, explicit non-event diagnostics, and a volatility-regime coverage audit that is not treated as headline evidence when intrawindow pseudo-event coverage is inadequate.

The asset-pricing channel is grounded in preferred-habitat and duration-risk theories. Greenwood and Vayanos (2014) show that the supply and maturity structure of government debt affect yields and expected returns through the price of duration risk. Krishnamurthy and Vissing-Jorgensen (2012) emphasize the liquidity and safety services of Treasuries, which means supply news can move spreads by changing the scarcity value of safe assets. Term-premium measurement follows Adrian et al. (2013) and the NY Fed ACM data.

The high-frequency identification design is related to monetary-policy event studies such as Gurkaynak et al. (2005) and Gertler and Karadi (2015). The latter is especially relevant because it uses high-frequency surprises as external instruments in a macro-financial VAR. The local-projection side builds on Jorda (2005), while the LP/VAR connection and finite-sample tradeoffs are guided by Plagborg-Moller and Wolf (2021). The lag-augmented inference logic follows Montiel Olea and Plagborg-Moller (2021).

The replication archive also records optional state-space, Monte Carlo, and spectral extensions without making them part of the core evidence. Kalman filtering and simulation smoothing (Carter and Kohn, 1994; Durbin and Koopman, 2002; Giannone et al., 2015) are useful for propagating measurement uncertainty in the multi-asset proxy, while frequency-domain tools can summarize transmission by horizon. These tools are deliberately not used as substitutes for the event-study identifying assumptions and are not reported as paper evidence until their diagnostics are complete.

3 Data

3.1 Treasury Events

The event database starts from Treasury Fiscal Data and TreasuryDirect auction records, which provide announcement dates, auction dates, security type, security term, offering amount, CUSIP, and auction-result variables. Quarterly Refunding Announcement documents from Treasury are the next priority because they contain broad debt-management news, Treasury Borrowing Advisory Committee materials, and primary-dealer expectation information.

The replication archive records timestamp provenance, source URLs, and whether each timestamp is exact or assumed. Date-only auction announcements are useful for source auditing and daily robustness, but they are not baseline high-frequency events until exact release times are verified.

3.2 High-Frequency Market Reactions

WRDS TAQ is the baseline market-reaction source. It provides intraday U.S. equity/ETF trades, so the baseline historical high-frequency proxy uses Treasury ETFs: `SHY`, `IEF`, `TLT`, `EDV`, `GOVT`, `BIL`, and `SHV`. Databento supplies a 2011–2026 one-minute futures robustness panel for continuous Treasury and rate futures. LSEG is used for news audit, Reuters story metadata, fiscal-confound headlines, and expectation/survey corroboration where available; it is not the baseline intraday market-data source unless Tick History/DataScope entitlement is added.

For each event e, the raw object is a minute-level VWAP panel

$$P_{k,e,\tau}, \qquad k \in \mathcal{K},$$

where $\mathcal{K}$ is the set of ETFs or futures and τ is event time in minutes.

The primary WRDS ETF windows are QRA $[-10, +30]$ and coupon announcement $[-5, +20]$. The canonical file collapses reactions to one observation per exact information-release timestamp. Security-level rows are retained for issuance quantities and expected-size matching, but the same ETF return is not counted once per security when several coupon securities are announced together. Legacy event-row shocks are retained only for audit and event-ID-based Databento comparisons. The serious-confound screen is event-type specific: QRA releases use a $[-60, +45]$-minute scheduled-confound window, while coupon announcements retain the $[-30, +60]$-minute screen; same-day scheduled releases outside those windows are retained as severity-one flags rather than excluded from the baseline. For validation, the same raw $[-30, +60]$ minute panel is also summarized into narrower windows, especially coupon $[-5, +15]$ and coupon $[-2, +10]$. These derived windows are not new data sources; they are alternative measurements of the same exact-minute information releases used to assess the bias-variance tradeoff between contamination risk and delayed price discovery.

For the matched non-event validation, the paper also constructs a daily volatility-regime panel from Cboe volatility indexes and FRED's $VXTYN$. The intended volatility-regime pseudo-events match Treasury events on weekday, clock time, lagged VIX tercile, and lagged rate-volatility tercile when those fields are available. Tercile cutoffs are expanding-window cutoffs formed from information available through the previous day, not full-sample cutoffs. This does not create a new shock series; in the finalized sample it is retained as a coverage audit because the accepted pseudo-event windows lack enough intrawindow trade coverage to produce non-mechanical returns.

The Databento futures panel is deliberately treated as a robustness sample. The paper screens 'ZT', 'ZF', 'ZN', 'ZB', and 'UB' as core Treasury futures, while 'TN', 'ZQ', and 'SR3' must pass coverage and activity thresholds before entering any paper specification. The futures proxy is never spliced into the WRDS ETF baseline, because the two panels measure related but not identical securities and the Databento sample starts only after the beginning of the baseline ETF sample.

3.3 Macro-Financial Outcomes

The monthly analysis panel merges FRED yields, credit spreads, activity variables, inflation, and financial-stress proxies with NY Fed ACM term-premium data and daily futures aggregates. The replication archive writes the merged panel and a coverage report.

The main outcomes are Treasury yield changes, ACM term-premium and risk-neutral yield changes, credit spreads, equity/volatility proxies, industrial production growth, payroll growth, investment growth, inflation, and unemployment changes. Yield and ACM changes are monthly changes in percentage-point units; multiplying by 100 gives basis points. ACM estimates are current-vintage New York Fed estimates rather than real-time vintage estimates, so the ACM decomposition is interpreted as a current-estimate decomposition of the yield response. ACM levels are retained as appendix robustness only because persistent level variables require additional stationarity and long-run interpretation checks.

For reference-month macro series such as industrial production, payrolls, PCE inflation, and unemployment, the $h = 0$ LP coefficient follows the standard reference-month convention: it pairs the Treasury proxy in month t with the macro value dated month t. These coefficients are not real-time macro announcement responses because the corresponding BLS/BEA releases are published with a delay.

Table 1: Main monthly outcome transformations.

Outcome family	Main transformation	Interpretation
Treasury yields	Monthly first difference	Percentage-point yield change
ACM components	Monthly first difference	Term-premium / risk-neutral yield change
Credit spreads	Level spread	Percentage-point spread
Industrial production, payrolls, PCE	Annualized log growth	Monthly growth rate at annual rate
Unemployment	Monthly first difference	Percentage-point unemployment change

3.4 Mechanism Data

The mechanism extensions use monthly state variables that are predetermined relative to the monthly Treasury supply-news proxy. The dealer-balance-sheet state comes from official NY Fed primary-dealer positions, financing, and fails data, supplemented by the OFR NYPD mirror. Positioning states come from CFTC Treasury futures reports. Rate-uncertainty states use Cboe Treasury ETF volatility indexes, FRED's historical $VXTYN$, and a Databento settlement-based Treasury option implied-volatility prototype. Corporate-credit transmission uses WRDS TRACE/FISD-derived monthly corporate-bond returns, yields, spreads, duration, and liquidity variables by rating bucket.

These mechanism variables are not high-frequency identification sources. They enter either as lagged state variables S_{t-1} in interaction local projections or as monthly transmission outcomes. State variables are standardized with expanding monthly moments and then lagged one month before entering the projection. This respects the filtration requirement: the state used to condition the effect of z_t must be known before the shock month and cannot be scaled with future observations.

3.5 Appendix-Only Extension Data

Several data layers are retained for appendix or future extensions rather than for the core claim. SEC N-PORT and WRDS CRSP report-date ETF holdings provide duration precision checks but not exact daily event-date holdings. Databento option data currently provide a daily settlement-based implied-volatility state panel rather than intraday event-window option-IV shocks. LSEG vendor rows and story bodies corroborate recent auction-size expectations, but they are not a complete historical tenor-level consensus panel. Databento futures definitions support robustness work, while exact CTD delivery-basket mapping remains deferred until CME Treasury Analytics, Bloomberg/LSEG CTD labels, or a local CTD computation is available.

4 Shock Construction

4.1 Release-Group Reactions

Let g index an information-release group, defined by an exact Treasury release timestamp. This distinction matters because many coupon announcement days contain several securities released at the same minute. The paper aggregates security-level quantities within g, but it counts the market reaction only once. For asset k and release group g, define the event-window log return

$$\Delta p_{k,g} = \log P_{k,g,\tau_1} - \log P_{k,g,\tau_0},$$

where τ_0 and τ_1 are pre- and post-announcement window endpoints. Because Treasury ETF prices move inversely with yields, the current price-based yield-oriented proxy is

$$\tilde{r}_{k,g} = -10{,}000\Delta p_{k,g}.$$

This sign convention makes positive values correspond to yield increases. It is not yet a duration-adjusted basis-point yield change.

The duration-adjusted target is

$$\Delta y_{k,g} \approx -\frac{\Delta P_{k,g}}{D_{k,g}P_{k,g}},$$

where $D_{k,g}$ is effective duration. This duration-adjusted version is used as an interpretation diagnostic; the raw release-window price-reaction proxy remains the baseline identified object.

4.2 Level and Long-Short Proxies

The level proxy is the average long/intermediate Treasury price reaction:

$$R_g^{LEV} = |\mathcal{L}_g|^{-1} \sum_{k \in \mathcal{L}_g} \tilde{r}_{k,g}, \qquad \mathcal{L} = \{\text{IEF}, \text{TLT}, \text{EDV}, \text{GOVT}\}.$$

Here $\mathcal{L}_g \subseteq \mathcal{L}$ is the set of level ETFs with usable quotes in release group g. The implementation therefore uses an available-ETF mean, not a fixed four-ETF mean; release groups with no usable level ETF are dropped from the level proxy. The raw long-short price-reaction proxy is

$$R_g^{SLOPE} = |\mathcal{H}_g|^{-1} \sum_{k \in \mathcal{H}_g} \tilde{r}_{k,g} - |\mathcal{S}_g|^{-1} \sum_{k \in \mathcal{S}_g} \tilde{r}_{k,g},$$

where $\mathcal{H} = \{\text{TLT}, \text{EDV}\}$ and $\mathcal{S} = \{\text{SHY}, \text{BIL}, \text{SHV}\}$. The sets $\mathcal{H}_g$ and $\mathcal{S}_g$ are defined analogously. This object is not automatically a maturity-composition shock. Long-duration ETF prices mechanically move more than short-duration ETF prices under a parallel yield shift. The paper therefore labels it as a raw long-short price-reaction proxy unless duration adjustment or orthogonalization supports a stronger interpretation.

Monthly raw proxies are sums over release groups:

$$Z_t^j = \sum_{g \in t} R_g^j, \qquad j \in \{LEV, SLOPE\}.$$

The standardized instrument used in local projections is

$$z_t^j = \frac{Z_t^j - \hat{\mu}_{j,t-1}}{\hat{\sigma}_{j,t-1}},$$

where $\hat{\mu}_{j,t-1}$ and $\hat{\sigma}_{j,t-1}$ are expanding moments computed from at least twelve prior nonmissing monthly observations. This point-in-time standardization avoids using future shock realizations to scale the proxy.

4.3 Interpretation

The level proxy is the baseline market-implied supply-news measure if it loads broadly across maturities. The raw long-short proxy is a candidate maturity-composition diagnostic only after accounting for duration exposure. The interpretation is checked empirically through release-window loadings, first-stage relationships with issuance quantities, duration-adjusted diagnostics, and robustness to asset sets.

4.4 Falsification Proxies

The identifying variation is expected to be concentrated at the Treasury release time, not in the minutes before the release or on matched non-event days. Placebo pre-window reactions are constructed using the same WRDS minute panel:

$$\tilde{r}_{k,g}^{pre(a,b)} = -10{,}000 \left(\log P_{k,g,b} - \log P_{k,g,a}\right), \qquad (a,b) \in \{(-30,-11), (-25,-6)\}.$$

The placebo level and slope proxies use the same ETF sets $\mathcal{L}, \mathcal{H}, \mathcal{S}$ as the baseline proxy. A credible announcement design requires these pre-window reactions to be weaker and less systematically related to the true event-window proxy.

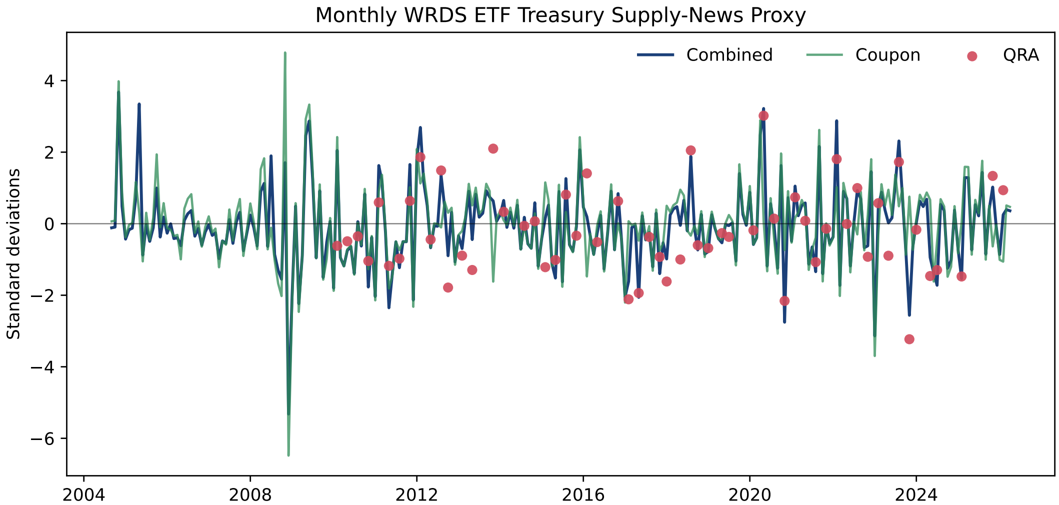

Figure 1: Monthly WRDS ETF Treasury supply-news proxy. The combined series is standardized after monthly aggregation; QRA and coupon components show how refunding and auction-announcement news enter the baseline proxy.

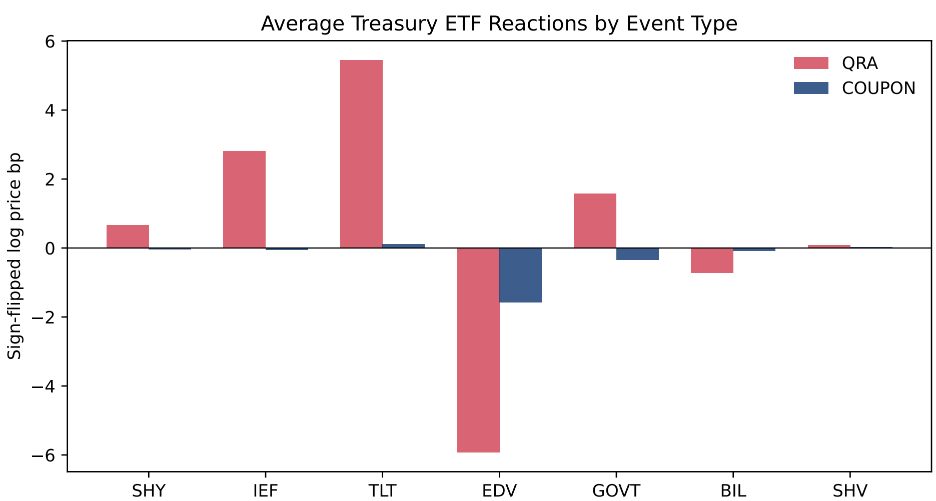

Figure 2: Average announcement-window ETF reactions. Values are sign-flipped log price basis points, $-10{,}000\Delta \log P$, not duration-adjusted yield basis points; positive bars mean ETF prices fall and yield pressure rises during Treasury release windows.

Matched non-event proxies are constructed at the same clock time and weekday as the original event, using candidate dates at ± 7 and ± 14 calendar days. Candidate pseudo-events are rejected if they overlap Treasury events or major scheduled fiscal, macro, or monetary-policy confounds within a narrow window. These matched windows test whether the baseline reactions are specific to Treasury information releases rather than normal intraday Treasury ETF volatility.

4.5 Narrow-Window Robustness Proxies

The matched non-event test shows that same-clock Treasury ETF volatility is large, so the analysis also constructs tighter announcement windows. The baseline uses QRA $[-10, +30]$ and coupon $[-5, +20]$. The narrow-window variants are

$$W_1 : \{\text{QRA } [-10, +30],\ \text{coupon } [-5, +15]\}, \qquad W_2 : \{\text{QRA } [-10, +30],\ \text{coupon } [-2, +10]\},$$

$$W_3 : \{\text{QRA } [-5, +15],\ \text{coupon } [-5, +15]\}, \qquad W_4 : \{\text{QRA } [-5, +15],\ \text{coupon } [-2, +10]\}.$$

For each variant, ticker-level returns are recomputed from the raw $[-30, +60]$ WRDS TAQ minute file, event-level level and slope proxies are rebuilt, monthly proxies are reaggregated, and the direct LPs are rerun. This is a bias-variance check: tighter windows reduce exposure to other news but may discard delayed price discovery. A robust baseline preserves signs and broad magnitudes under W_1, while W_2 is a conservative stress test rather than a preferred specification.

4.6 Event-Type and Confound Decompositions

The combined raw monthly shock can be decomposed as

$$Z_t^{LEV} = Z_{t,QRA}^{LEV} + Z_{t,coupon}^{LEV},$$

where QRA release groups and coupon auction announcement release groups are aggregated separately. The combined series is then standardized for the main LP. Separately standardized QRA-only and coupon-only diagnostics do not add up mechanically to the combined standardized proxy. This split checks whether refunding-policy news and ordinary coupon announcement news have similar transmission. A stricter severity-zero raw proxy is also constructed:

$$Z_{t,sev0}^{LEV} = \sum_{g \in t:\, severity_g = 0} R_g^{LEV},$$

which excludes all events with any known scheduled or news confound. The main baseline remains the severity 0–1 sample, while severity-zero results are the conservative exclusion-robustness check.

5 Identification

5.1 External-Instrument Restrictions

Let s_t be the latent Treasury supply-news component of interest and z_t be the high-frequency market-implied proxy. The external-instrument restrictions that would justify a stronger LP-IV or proxy-SVAR interpretation are

$$\mathbb{E}[z_t s_t] \neq 0, \qquad \mathbb{E}[z_t \varepsilon_{-s,t}] = 0.$$

The first condition is relevance. The second is exclusion: within the announcement window, the instrument must not primarily measure non-Treasury news such as monetary-policy news, macro data releases, or unrelated market stress. The direct-LP baseline does not claim to observe s_t itself; it estimates responses to the standardized proxy z_t.

The implemented LPs condition on lags and predetermined controls W_{t-1}, so the empirical restrictions are best written after Frisch-Waugh residualization. Let M_W denote residualization with respect to the control set and define

$$\tilde{z}_t = M_W z_t, \qquad \tilde{x}_t = M_W x_t, \qquad \tilde{y}_{i,t+h} = M_W y_{i,t+h}.$$

Then relevance for an observed supply quantity x_t is $\mathrm{Cov}(\tilde{z}_t, \tilde{x}_t) \neq 0$, and the stronger exclusion interpretation requires $\mathrm{Cov}(\tilde{z}_t, \tilde{\varepsilon}_{-s,t}) = 0$. The lead-lag exclusion condition is stronger: the proxy should not systematically forecast pre-announcement outcomes or unrelated shocks once W_{t-1} is controlled for. The placebo, matched non-event, predictability, and confound checks are designed to probe these implications, not to prove them mechanically.

For multiple shocks,

$$s_t = (s_t^{VOL}, s_t^{MAT})', \qquad z_t = (z_t^{LEV}, z_t^{SLOPE})',$$

identification requires the relevance matrix

$$\Pi = \mathbb{E}[z_t s_t']$$

to have full rank after the same conditioning, and the instruments must be excluded from non-supply news.

5.2 LP-IV Estimand

If x_t is an observed supply variable, such as a duration-equivalent issuance surprise, first stage and second stage are

$$x_t = \pi z_t + \Gamma' W_{t-1} + v_t,$$
$$y_{i,t+h} = \alpha_{i,h} + \beta_{i,h}\hat{x}_t + \Theta_{i,h}' W_{t-1} + u_{i,t+h}.$$

The scalar IV estimand is

$$\beta_{i,h}^{IV} = \frac{\mathrm{Cov}(\tilde{z}_t, \tilde{y}_{i,t+h})}{\mathrm{Cov}(\tilde{z}_t, \tilde{x}_t)}.$$

If x_t is not yet credible, the direct LP

$$y_{i,t+h} = \alpha_{i,h} + \theta_{i,h} z_t + \Theta_{i,h}' W_{t-1} + u_{i,t+h}$$

is the baseline reduced-form response to a standardized Treasury supply-news proxy.

5.3 Falsification Logic

The exclusion restriction is not testable directly, but it has empirical implications. First, placebo pre-window proxies are expected to be weaker than true announcement-window proxies:

$$\mathbb{E}\left[|z_t^{pre}|\right] < \mathbb{E}\left[|z_t|\right], \qquad \mathrm{Cov}(z_t^{pre}, z_t) \text{ small}.$$

Second, matched non-event windows are expected to be smaller and less systematically related to the true announcement-window reactions. This is a demanding falsification test because Treasury ETFs and futures can move for ordinary market-liquidity, rate-news, and risk-sentiment reasons even when no Treasury release occurs. Therefore, a large non-event absolute reaction weakens the signal-to-noise interpretation of the proxy, while a low matched-event correlation and larger true-event reaction still provide evidence that the announcement window contains event-specific information. Third, the proxy is expected not to be strongly predictable from lagged public information W_{t-1}:

$$z_t = a + \Gamma' W_{t-1} + u_t, \qquad H_0 : \Gamma = 0.$$

These tests do not prove exogeneity, but failure would directly weaken the maintained assumption $E[z_t \varepsilon_{-s,t}] = 0$. This is why the paper's main language is "responses to a market-implied Treasury supply-news proxy" rather than an unqualified structural-shock claim.

The volatility-regime matched non-event design is retained as an audit of the second falsification test rather than as a headline placebo result in the finalized sample. Calendar-matched pseudo-events can overstate the placebo challenge if they fall in different volatility regimes from the Treasury event, so the intended regime-matched design holds fixed lagged VIX and rate-volatility terciles when usable intrawindow pseudo-event data exist. Under the finalized WRDS pull, however, the accepted regime-matched pseudo-event summaries have only one minute per accepted ticker-window, which would make non-event returns mechanically zero. The paper therefore appendix-gates this extension and uses the calendar-matched non-event diagnostic as the valid matched-window evidence.

Narrow-window robustness is a complementary falsification design. If the baseline response is mainly unrelated market volatility accumulated over too wide a window, then recomputing the proxy with coupon $[-5, +15]$ or $[-2, +10]$ windows would materially change the monthly proxy and overturn the direct-LP signs. If instead the narrower proxy remains highly correlated with the baseline and preserves the main yield responses, the evidence is consistent with a noisy but informative market-implied supply-news proxy. This does not transform z_t into an observed structural shock; it strengthens the measurement and exclusion argument by showing that the baseline is not mechanically driven by late-window contamination.

Relevance is assessed at two levels. Market relevance is visible in event-window ETF reactions and their Databento futures robustness. Quantity relevance is assessed by regressions and first stages that link the market shock to realized and expected issuance surprises. Quantity relevance is necessary before any LP-IV estimate can be interpreted as the response to a scaled supply quantity rather than to the market-implied news proxy itself.

5.4 Recoverability and FEVD Guardrail

External instruments identify impulse responses under weaker conditions than those needed for conventional forecast-error variance decompositions or historical decompositions. Therefore, this paper reports direct LP responses as the baseline and LP-IV only as secondary evidence. It reports FEVD/HD only under explicit recoverability and structural-completeness assumptions, and otherwise treats coherency and transfer-function objects as appendix diagnostics.

6 Estimation and Inference

6.1 Lag-Augmented Local Projections

The baseline local projection uses the same predetermined-control notation as the identification section:

$$y_{i,t+h} = \alpha_{i,h} + \theta_{i,h} z_t + \Theta'_{i,h} W^{(p)}_{i,t-1} + u_{i,t+h}.$$

Here $W_{i,t-1}^{(p)}$ is the lag-expanded predetermined information set used for outcome i. For each month t, the baseline matrix contains the lagged shock, lagged outcome, and lagged macro-financial controls:

$$W_{i,t-1}^{(p)} = \left(z_{t-1}, \ldots, z_{t-p},\ y_{i,t-1}, \ldots, y_{i,t-p},\ q'_{t-1}, \ldots, q'_{t-p}\right)',$$

where q_t contains the macro-financial controls listed in Table 2, and $\Theta_{i,h}$ stacks the corresponding lag coefficients. Thus the baseline $p = 3$ specification includes three lags of each listed variable and no contemporaneous macro-financial controls. Factor-control robustness appends three or five fixed-loading macro-financial factors to q_t before forming the same lag-expanded $W_{i,t-1}^{(p)}$.

Table 2: Variables entering the predetermined control set $W_{i,t-1}^{(p)}$.

Variable	Interpretation
z_t	Expanding-standardized market-implied Treasury supply-news proxy; included through lags $z_{t-1}, \ldots, z_{t-p}$.
$y_{i,t}$	The outcome in the current LP equation; included through its own lags $y_{i,t-1}, \ldots, y_{i,t-p}$.
`DGS2`	FRED two-year Treasury constant maturity yield level.
`DGS10`	FRED ten-year Treasury constant maturity yield level.
`term_spread_10y2y`	Ten-year minus two-year Treasury yield spread, `DGS10-DGS2`.
`VIXCLS`	Cboe VIX index level from FRED, used as a broad equity-market uncertainty control.
`baa_aaa_spread`	Moody's seasoned Baa corporate bond yield minus Moody's seasoned Aaa corporate bond yield; a corporate-credit risk-spread control.
`INDPRO`	FRED industrial production index level.
`UNRATE`	FRED civilian unemployment rate.
`mf_factor_1--5`	Fixed-loading macro-financial PCA factors used only in factor-control robustness specifications.

This choice follows the lag-augmented LP logic in Montiel Olea and Plagborg-Moller (2021): including predetermined lags makes the baseline finite-sample-corrected Eicker-Huber-White inference less sensitive to persistence and long-horizon overlap than naive autoregressive delta-method inference. The $h = 0$ coefficient is a same-month monthly response to the high-frequency proxy, not a release-window yield response. Release-window reactions are measured directly only for ETF and futures prices.

6.2 Standard Errors

The implementation computes HC1 Eicker-Huber-White standard errors after Frisch-Waugh residualization. The HC1 correction multiplies the sandwich covariance by $n/(n - k)$, where k is the number of regressors in the residualized LP including the intercept and lag controls. Reported 90 percent intervals use Student-t critical values with the corresponding residual degrees of freedom. HAC/Newey-West covariance estimates are reserved for non-augmented or explicitly serial-correlation-sensitive specifications. This distinction matters because, in lag-augmented LPs, EHW scores can be valid even though projection residuals are serially correlated.

6.3 Sensitivity Designs

The paper reports the three-lag EHW direct LP as the baseline. Sensitivity checks vary the lag length $p \in \{1, 2, 3, 6, 12\}$, the proxy definition $z_t \in \{z_t^{LEV}, z_{t,sev0}^{LEV}, z_{t,QRA}^{LEV}, z_{t,coupon}^{LEV}\}$, and the control set. The narrow-window robustness checks then rebuild z_t^{LEV} from tighter event-window summaries, including coupon $[-5, +15]$ and coupon $[-2, +10]$, and rerun the same LP at the baseline peak horizons. These alternative windows are measurement robustness checks, not separate economic shocks. Factor-augmented controls use fixed PCA loadings estimated before the high-frequency proxy sample begins, following the high-dimensional factor-model motivation in Stock and Watson (2016). These factors are controls, not structural shocks.

For inference robustness, Newey-West/Bartlett standard errors use bandwidth $\max\{h, 3\}$, following long-run variance logic and Newey and West (1987). The replication output records the rank and condition number of the Newey-West bread matrix; the implementation uses a pseudo-inverse only as numerical protection. Block-bootstrap intervals use circular monthly blocks of length 6 and 12 with fixed random seed over the Frisch-Waugh residualized y and z series. This is a residualized circular-block bootstrap robustness check, not a full joint block bootstrap of the entire lagged panel. The bootstrap is reported only as robustness because the time-series bootstrap requires extra dependence assumptions and can fail in finite samples.

6.4 State-Dependent Mechanism Projections

The mechanism extensions estimate interaction local projections of the form

$$y_{t+h} = \alpha_h + \theta_h z_t + \delta_h z_t S_{t-1} + \phi_h S_{t-1} + \Gamma_h' W_{t-1} + u_{t+h}.$$

Here S_{t-1} is an expanding-standardized dealer-balance-sheet, positioning, or rate-uncertainty state. The construction is deliberately point-in-time. For any raw monthly state component a_t, define

$$\mathcal{Z}_t(a) = \frac{a_t - \bar{a}_{1:t}}{s_{1:t}}, \qquad \bar{a}_{1:t} = \frac{1}{t}\sum_{\tau \le t} a_\tau,$$

with at least twelve nonmissing monthly observations required before a standardized value is used. The regression for month t then uses S_{t-1}, the previous month's standardized state. Equivalently, the state can use information through month $t-1$, but it never uses month-t outcomes or full-sample scaling. This one-month lag is essential: it keeps the state inside the information set before the shock and prevents the mechanism regression from conditioning on variables partly determined by the announcement itself.

Table 3: Construction of state variables used in S_{t-1}.

State variable	Construction before the one-month lag
Rate uncertainty	Row-wise average of $\mathcal{Z}_t$(`VXTLT`), $\mathcal{Z}_t$(`VXIEF`), and $\mathcal{Z}_t$(`VXTYN`), using monthly last observations. `VXTLT` and `VXIEF` are Cboe Treasury-ETF option-volatility indexes; `VXTYN` is FRED's discontinued ten-year Treasury-note volatility index.
Dealer constraint	Average of expanding-standardized NY Fed/OFR primary-dealer balance-sheet components: log(1+Treasury fails), log(1+Treasury financing), log(1+absolute long coupon position), and the fails-to-position ratio.
CFTC dealer position	$\mathcal{Z}_t$ of CFTC dealer/intermediary duration-weighted net Treasury futures positioning scaled by open interest.
CFTC leveraged money	$\mathcal{Z}_t$ of CFTC leveraged-fund duration-weighted net Treasury futures positioning scaled by open interest.

Following the nonlinear-dynamic-effects cautions in Kolesár and Plagborg-Møller (2025), δ_h is interpreted as a conditional projection effect. It is not a new source of exogeneity and it is not, by itself, a fully structural nonlinear impulse response. A mechanism result is promoted only when the interaction sample is large enough, the sign is stable under factor controls, and the result sharpens the economic argument.

6.5 Weak-IV Robust Intervals

For any proposed β_0, the Anderson-Rubin regression is

$$y_{i,t+h} - \beta_0 x_t = a + \Gamma' W_{t-1} + \delta z_t + e_t.$$

The weak-IV robust confidence set is

$$CI_{AR} = \{\beta_0 : \text{do not reject } H_0 : \delta = 0\}.$$

The code implements this grid-inversion logic when a valid endogenous supply quantity and instrument are supplied.

LP-IV is not the main specification. It is run only for the combined expected-size surprise after the first stage passes the relevance rule. Anderson-Rubin intervals, following Anderson and Rubin (1949) and the weak-IV cautions summarized by Andrews et al. (2019), determine whether the LP-IV magnitudes can be reported as secondary evidence. If the first stage or AR intervals are unstable, the LP-IV results are appendix-only or deferred.

6.6 Bootstrap

The bootstrap is not the default for the baseline LP because the statistical notes stress that bootstrap validity depends on the estimator and dependence structure. It is useful later for max-t joint bands, block-bootstrap robustness, and VAR/BVAR comparison, but not as a substitute for the analytical LP inference argument.

7 Validation and Robustness

The validation strategy follows the external-instrument logic used in high-frequency macro-finance designs. A high-frequency proxy can support causal interpretation only if the timing is exact, the release window is not contaminated by other news, the proxy is relevant for Treasury supply news, and the measured responses are not artifacts of normal market volatility. The main text therefore reports the matched non-event diagnostic and the primary narrow-window robustness table. The appendix contains the full placebo, QRA/coupon, severity, Databento, lag-sensitivity, and first-stage tables.

7.1 Shock Validity Tests

Appendix Table 11 reports predictability and serial-dependence diagnostics for the baseline monthly proxy. The key falsification tests are placebo pre-windows and matched non-event windows. Appendix Table 12 compares the mean absolute pre-announcement reactions with true event-window reactions. Table 4 reports the main matched non-event WRDS pull using the same weekday and clock time as each Treasury event, with QRA $[-10, +30]$ and coupon $[-5, +20]$ primary windows.

Matched non-event windows are large. After collapsing to information-release groups, the matched pseudo-event sample covers 550 release groups and 7,550 ticker-window observations.

Mean absolute reactions equal 8.61, compared with 10.35 in true release windows; the paired difference is statistically different from zero, but the pseudo-event reaction is 83.2 percent as large as the event reaction. Because true and matched reactions are essentially uncorrelated, the test supports event-specific signed variation while showing that normal same-clock Treasury volatility is quantitatively important.

The volatility-regime matched extension is retained as an appendix-gated diagnostic rather than a headline result. It matches pseudo-events on same weekday, same clock time, lagged VIX tercile, and lagged rate-volatility tercile when available, using expanding-window tercile cutoffs based only on information available through the previous day. In the finalized sample, however, the reusable WRDS pseudo-event windows have insufficient intrawindow trade coverage: the primary summaries contain only one minute per accepted ticker-window, so the non-event return would be mechanically zero. The paper therefore does not report the volatility-regime matched row as evidence; the calendar-matched non-event pull remains the valid matched-window diagnostic.

This result disciplines the interpretation. The analysis does not claim that all movement inside the event window is Treasury-release news, and the matched non-event test is not treated as proof of exclusion. The defensible claim is narrower: official Treasury release windows contain larger Treasury-market repricing than matched non-event windows, but residual non-event volatility requires careful confound screens and robustness to narrower windows or severity-zero events.

Table 4: Matched non-event diagnostics. The table compares absolute release-window reactions with pseudo-event reactions at the same clock time.

Design	Release groups	Ticker-window rows	True abs. reaction	Matched abs. reaction	Matched / true	True / matched	Share true larger	Signed corr.
Calendar matched	550	7550	10.3506	8.6112	0.8320	1.2020	0.5273	-0.0255

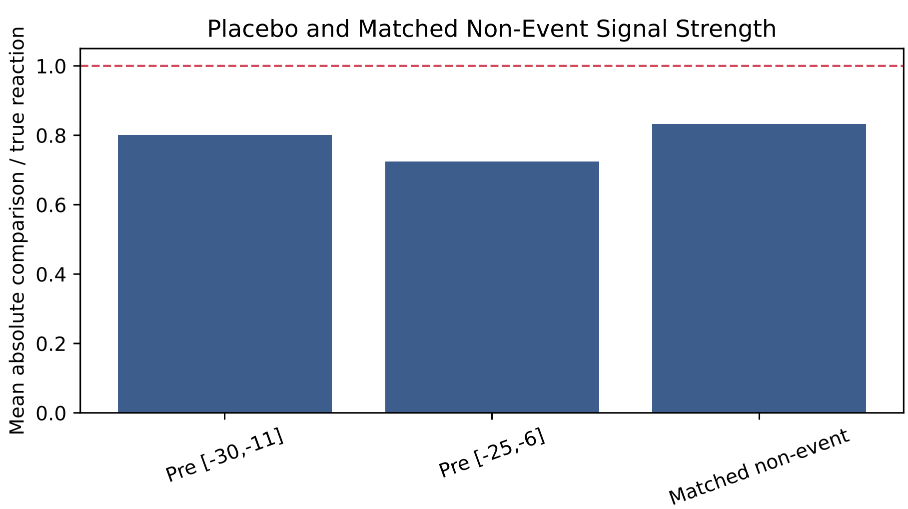

Figure 3: Placebo and matched non-event reaction strength relative to true Treasury announcement windows. Values near one indicate that ordinary same-clock market volatility is large relative to the announcement-window proxy.

7.2 Event-Type and Confound Robustness

Table 14 separates QRA and coupon-announcement proxies in the appendix. This matters because QRA news is broad debt-management policy news, while coupon announcements can be more

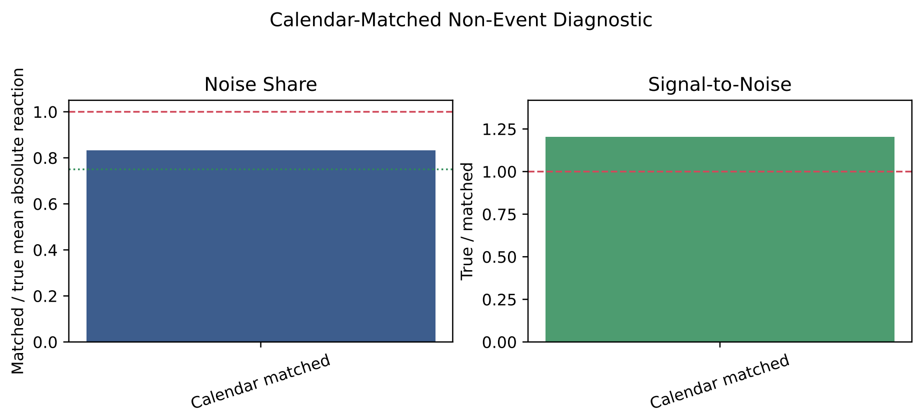

Figure 4: Matched non-event diagnostics. The current volatility-regime extension is appendix-gated because the reusable pseudo-event windows have insufficient intrawindow trade coverage.

tenor-specific and mechanical. Table 15 compares the baseline severity 0–1 proxy with the stricter severity-zero proxy and the wide-window robustness proxy. QRA releases motivate the design because they are closest to broad issuance-policy news, while coupon announcements provide most of the monthly empirical power because they occur more often.

7.3 Influence and Issuance Relevance

The influence diagnostics identify the largest event and month contributions, reconstruct winsorized monthly proxies, and run leave-one-month-out checks at the baseline peak horizons. These checks are not meant to remove influential events automatically; they identify whether the main conclusions are broad or hinge on a small number of episodes.

The realized-issuance relevance tests link market reactions to expected-size surprises, issuance changes, and duration-equivalent surprises. They are interpretation and relevance evidence. They do not prove exclusion, and they do not replace the high-frequency market-implied proxy as the baseline object.

7.4 Direct-LP Robustness

The main direct LP uses three lags and HC1 Eicker-Huber-White standard errors. Table 18 reports peak-horizon sensitivity to lag length, factor controls, and severity-zero shock construction. HAC and residualized circular-block-bootstrap intervals are computed in the replication output as inference robustness checks after the baseline HC1 EHW results; the main appendix reports the compact lag and control sensitivity table.

The lag and control sensitivity checks preserve the qualitative signs of the main descriptive peak responses. Across $p \in \{1, 2, 3, 6, 12\}$, the 10-year same-month response is tightly clustered between about 3.80 and 4.07 basis points, while the 2-year same-month response ranges from about 2.87 to 3.81 basis points. Adding factor controls or using the severity-zero shock leaves the same-month yield responses positive, while the 5-year descriptive reversal remains negative at the baseline peak horizon. The PCE inflation response also remains positive across these specifications. The macro activity responses keep the expected signs–lower industrial production and payroll growth, higher unemployment–but remain imprecise, so the robustness evidence is about sign discipline rather than sharp macro magnitudes.

The main sensitivity caveat is the ACM decomposition. The selected-horizon same-month risk-neutral component is sharper than the term-premium component, while the descriptive peak term-premium response changes sign over the horizon range. The selected-horizon HAC and residualized circular-block-bootstrap checks leave the same-month yield responses statistically positive under Newey-West and both block lengths, but longer-horizon and macro intervals widen. For that reason, the three-lag HC1 EHW LP remains the main specification, while HAC and bootstrap intervals are inference robustness checks, and long-horizon macro conclusions are stated conservatively.

The full lag-sensitivity table is reported in Appendix Table 18.

7.5 Narrow-Window Robustness

Table 5 reports shock correlations after rebuilding the WRDS ETF reactions with tighter announcement windows. The main narrow coupon specification, QRA $[-10, +30]$ and coupon $[-5, +15]$, remains close to the baseline: the monthly level-shock correlation is 0.944, the coupon release-group correlation is 0.925, and the average absolute release-group reaction is about 94 percent of the baseline-window reaction. This supports using the baseline window while acknowledging that a few post-announcement minutes contribute to measured price discovery.

The most aggressive coupon window, QRA $[-10, +30]$ and coupon $[-2, +10]$, is a stress test. Its monthly level-shock correlation falls to 0.748, and the average absolute release-group reaction falls to about 84 percent of baseline. The attenuation is informative rather than disqualifying: tighter windows reduce exposure to ordinary market volatility but can miss slower absorption of Treasury supply news. The preferred narrow-window robustness check is therefore $[-5, +15]$, while $[-2, +10]$ is a conservative lower-signal specification.

Table 19 compares direct-LP coefficients at the baseline descriptive peak horizons. The yield responses keep the same sign under all narrow-window variants. For example, the 10-year same-month response remains positive under the coupon $[-5, +15]$ and coupon $[-2, +10]$ variants, with estimates around 3.5 and 2.0 basis points compared with the baseline 4.0 basis points. The 5-year descriptive reversal remains negative under all narrow-window variants. The term-premium and credit-spread estimates are more sensitive, so those channels remain supportive mechanisms rather than exact magnitude claims.

Table 5: Narrow-window proxy robustness diagnostics. Coupon [-5,+15] is the primary tight-window check; coupon [-2,+10] is a lower-signal stress test.

variant	description	event_count	monthly_z_level_corr_with_baseline	coupon_event_level_corr_with_baseline	qra_event_level_corr_with_baseline	variant_to_baseline_mean_abs_ratio	mean_abs_event_difference_from_baseline
coupon_5_15	QRA [-10,+30], coupon [-5,+15]	615	0.9443	0.9250	1.0000	0.9395	3.1253
coupon_2_10	QRA [-10,+30], coupon [-2,+10]	615	0.7480	0.7038	1.0000	0.8412	5.3509
all_5_15	QRA [-5,+15], coupon [-5,+15]	615	0.9017	0.9250	0.8526	0.9265	4.2130
qra_5_15_coupon_2_10	QRA [-5,+15], coupon [-2,+10]	615	0.6889	0.7038	0.8526	0.8283	6.4386

The corresponding narrow-window LP table is reported in Appendix Table 19. It is a sensitivity check on the direct-LP results, not a separate headline result.

7.6 Databento Futures Robustness

Databento futures provide an independent 2011–2026 robustness panel. Appendix Tables 21–24 screen continuous futures symbols, report WRDS/Databento overlap correlations, list disagreement events, and estimate futures-proxy LP robustness. These diagnostics are used to assess whether futures-market reactions tell the same broad story as ETF reactions, not to change the baseline proxy. Because Databento does not supply CTD labels in the current pipeline, these futures responses are raw price-reaction robustness checks rather than CTD-adjusted yield-equivalent estimates.

7.7 LP-IV Decision Rule

LP-IV is secondary. Table 7 in the results section reports whether the combined expected-size surprise passes the first-stage and Anderson-Rubin requirements needed for secondary interpretation. Survey-only surprises remain small-sample evidence, and TBAC recommendations and previous-size fallbacks are not market expectations and remain diagnostic unless separately justified.

8 Baseline Results

The main empirical object is the response to a one-unit expanding-standardized WRDS ETF Treasury supply-news proxy z_t^{LEV}. The coefficient is a reduced-form response to market-implied supply news, not a mechanically scaled dollar issuance shock. The dollar interpretation enters through expected-size and realized-issuance relevance diagnostics, and LP-IV is reported only after the direct LP evidence.

The main text is intentionally selective. It reports the WRDS ETF direct-LP responses, the matched non-event caveat, the primary narrow-window robustness check, the combined-surprise LP-IV decision table, and a compact mechanism table. The appendix preserves the audit trail: selected-horizon LPs, factor controls, lag sensitivity, QRA/coupon and severity splits, Databento futures robustness, full first-stage diagnostics, and full mechanism-decision rows. This hierarchy keeps the central claim clear: the paper estimates responses to a market-implied Treasury supply-news proxy, while LP-IV, futures, and mechanisms are secondary interpretation layers.

8.1 Direct Local Projections

Table 6 reports pre-specified monthly horizons $h \in \{0, 3, 6, 12, 24\}$. The coefficient is the response to a one-unit expanding-standardized high-frequency Treasury supply-news proxy, where the standardizing mean and variance are computed from prior monthly observations only. For yield and ACM outcomes, the table entries are in percentage-point units, so multiplying by 100 gives basis points. The descriptive peak-response table is kept in the appendix because it selects horizons after looking across $0, \ldots, 24$ and therefore does not provide valid peak-adjusted confidence intervals.

The first result is a positive same-month yield response. In the shock month, the 2-year, 5-year, 10-year, and 30-year monthly yield changes rise by about 3.2, 4.1, 4.0, and 3.9 basis points, respectively, with 90 percent HC1 Eicker-Huber-White intervals excluding zero. Given the matched non-event evidence in Section 7, this is a monthly response to release-window Treasury-market repricing associated with a market-implied Treasury supply-news proxy, not a perfectly isolated structural innovation. At horizon 12, the 2-year and 5-year monthly yield-change responses are negative and precisely estimated, while the 10-year and 30-year estimates are negative but less precise. These later negative coefficients mean that future monthly yield changes turn negative; cumulative yield-level effects would require summing the horizon coefficients and are not the entries in Table 6. The reduced-form dynamics may combine policy feedback, macro expectations, subsequent issuance information, and mean reversion in risk premia.

The ACM decomposition is more mixed than the raw yield response. Using monthly changes from current-vintage ACM estimates, the 10-year ACM term-premium response is about 0.6 basis points in the shock month, with a 90 percent interval that includes zero, while the 10-year ACM risk-neutral yield change is about 1.6 basis points and statistically sharper in the shock month. The descriptive peak table selects a later negative term-premium response, so the current estimates do not provide a clean positive term-premium channel. The baseline evidence is therefore consistent with Treasury supply news moving the yield curve, but the current-vintage ACM decomposition does not isolate whether the same-month response is primarily term-premium or expected-rate news.

The evidence outside Treasury markets is weaker. The same-month BAA–AAA spread response is negative and estimated precisely in the selected-horizon table, but the broader credit-spread path is not as stable as the Treasury-yield response. Industrial production and payroll growth have negative descriptive peak responses, and unemployment has a positive descriptive peak response, but these macro responses are imprecise at conventional confidence levels. For reference-month macro series, the $h = 0$ coefficient is not a real-time macro announcement response because those data are released with a delay. PCE inflation has a positive post-selected short-run response at horizon 1 with a 90 percent pointwise interval excluding zero, but this is descriptive because the horizon is selected from the peak-response table. The macro and credit estimates are therefore directional evidence to be checked against robustness and mechanism tests, not the core empirical contribution.

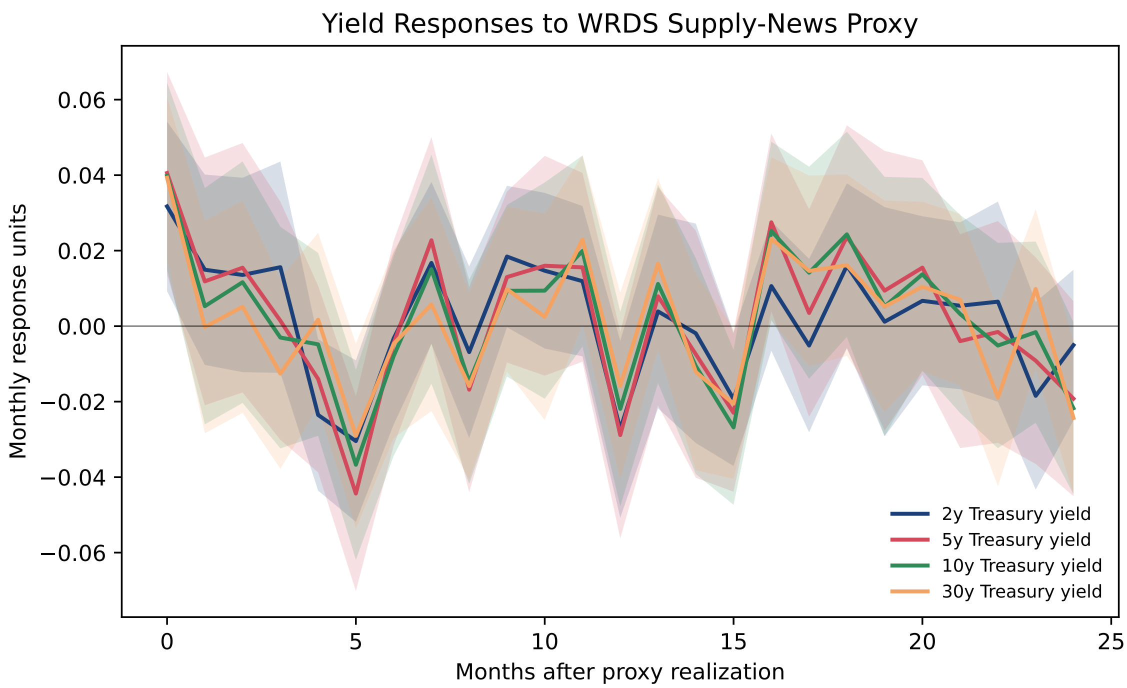

Figure 5: Monthly yield responses to the baseline WRDS Treasury supply-news proxy. Shaded bands are 90 percent HC1 Eicker-Huber-White intervals from the three-lag direct LP; these are monthly responses, not release-window yield changes.

8.2 Factor-Augmented Robustness

The factor-augmented results ask whether the direct LP response survives a richer predetermined macro-financial information set. Appendix Table 17 compares the baseline response at the baseline peak horizon with the factor3 specification. Large sign changes here would indicate that the baseline proxy is correlated with omitted macro-financial state variables; stable signs support the reduced-form interpretation.

The main yield responses are stable. At the baseline selected and descriptive peak horizons, the factor3 responses remain close to the baseline estimates: the 10-year same-month response

Table 6: Pre-specified selected-horizon direct local-projection responses to a one-unit expanding-standardized monthly WRDS Treasury supply-news proxy realization. Intervals use finite-sample-corrected Eicker-Huber-White (HC1) standard errors and Student-t critical values.

Outcome	Horizon	Response	HC1 EHW SE	90% CI lower	90% CI upper	N
2y Treasury yield	0	0.0317	0.0136	0.0092	0.0542	250
2y Treasury yield	3	0.0156	0.0169	-0.0124	0.0436	248
2y Treasury yield	6	-0.0033	0.0138	-0.0261	0.0195	247
2y Treasury yield	12	-0.0273	0.0142	-0.0508	-0.0039	242
2y Treasury yield	24	-0.0050	0.0121	-0.0250	0.0149	230
5y Treasury yield	0	0.0406	0.0162	0.0137	0.0674	250
5y Treasury yield	3	0.0014	0.0191	-0.0301	0.0330	248
5y Treasury yield	6	-0.0045	0.0164	-0.0316	0.0226	247
5y Treasury yield	12	-0.0289	0.0166	-0.0562	-0.0015	242
5y Treasury yield	24	-0.0192	0.0156	-0.0450	0.0066	230
10y Treasury yield	0	0.0398	0.0150	0.0151	0.0645	250
10y Treasury yield	3	-0.0030	0.0178	-0.0324	0.0263	248
10y Treasury yield	6	-0.0079	0.0160	-0.0343	0.0185	247
10y Treasury yield	12	-0.0219	0.0156	-0.0477	0.0038	242
10y Treasury yield	24	-0.0218	0.0138	-0.0445	0.0010	230
30y Treasury yield	0	0.0393	0.0128	0.0181	0.0605	250
30y Treasury yield	3	-0.0126	0.0153	-0.0378	0.0126	248
30y Treasury yield	6	-0.0047	0.0151	-0.0296	0.0203	247
30y Treasury yield	12	-0.0158	0.0150	-0.0406	0.0089	242
30y Treasury yield	24	-0.0243	0.0123	-0.0446	-0.0040	230
10y ACM term-premium change	0	0.0130	0.0145	-0.0109	0.0369	250
10y ACM term-premium change	3	-0.0028	0.0182	-0.0329	0.0274	247
10y ACM term-premium change	6	-0.0161	0.0188	-0.0472	0.0150	247
10y ACM term-premium change	12	0.0228	0.0150	-0.0020	0.0475	241
10y ACM term-premium change	24	0.0054	0.0153	-0.0199	0.0307	229
10y ACM risk-neutral yield change	0	0.0138	0.0087	-0.0006	0.0283	250
10y ACM risk-neutral yield change	3	-0.0141	0.0086	-0.0283	0.0002	247
10y ACM risk-neutral yield change	6	0.0085	0.0108	-0.0094	0.0263	247
10y ACM risk-neutral yield change	12	-0.0121	0.0096	-0.0279	0.0037	241
10y ACM risk-neutral yield change	24	0.0060	0.0086	-0.0082	0.0202	229
BAA-AAA spread	0	-0.0139	0.0086	-0.0282	0.0004	250
BAA-AAA spread	3	-0.0313	0.0196	-0.0637	0.0011	247
BAA-AAA spread	6	0.0076	0.0252	-0.0341	0.0493	247
BAA-AAA spread	12	0.0117	0.0198	-0.0210	0.0443	241
BAA-AAA spread	24	-0.0255	0.0200	-0.0585	0.0076	229
Industrial production growth	0	-0.7631	0.9686	-2.3629	0.8368	249
Industrial production growth	3	0.7638	0.8532	-0.6456	2.1732	247
Industrial production growth	6	-0.2106	0.9859	-1.8391	1.4180	246
Industrial production growth	12	0.7369	0.5437	-0.1613	1.6350	240
Industrial production growth	24	-0.3366	0.9986	-1.9869	1.3137	228
Payroll growth	0	-1.3188	0.9852	-2.9462	0.3086	250
Payroll growth	3	-0.4517	0.6072	-1.4548	0.5514	247
Payroll growth	6	-0.7093	0.8934	-2.1851	0.7665	247
Payroll growth	12	0.4807	0.5640	-0.4511	1.4124	241
Payroll growth	24	-0.6532	0.9350	-2.1984	0.8919	229
PCE inflation	0	-0.0445	0.1423	-0.2796	0.1906	249
PCE inflation	3	-0.1145	0.1577	-0.3750	0.1459	247
PCE inflation	6	-0.1969	0.1911	-0.5126	0.1189	246
PCE inflation	12	-0.0290	0.1365	-0.2545	0.1964	240
PCE inflation	24	0.0070	0.1225	-0.1955	0.2095	228
Unemployment rate change	0	0.0670	0.0605	-0.0330	0.1670	248
Unemployment rate change	3	0.0289	0.0360	-0.0306	0.0884	245
Unemployment rate change	6	0.0487	0.0577	-0.0466	0.1440	245
Unemployment rate change	12	-0.0322	0.0303	-0.0822	0.0179	239
Unemployment rate change	24	0.0473	0.0601	-0.0520	0.1466	227

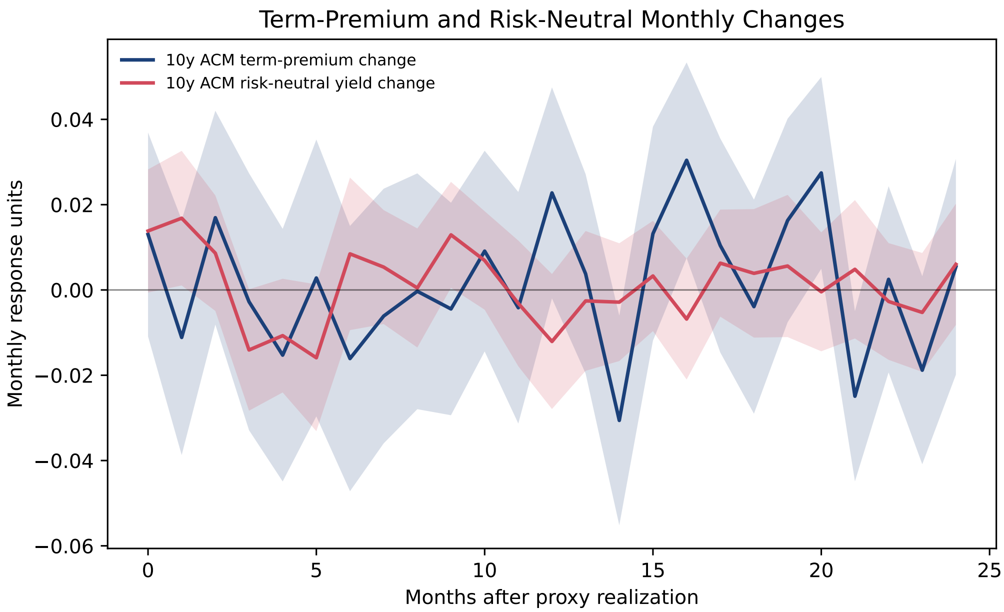

Figure 6: ACM decomposition using monthly changes in the 10-year term-premium and risk-neutral components from current-vintage estimates. The decomposition is useful for interpretation, but it does not isolate a clean positive term-premium channel in the current baseline.

stays near 4 basis points, and the 5-year descriptive reversal remains negative. The PCE inflation response and the directional macro responses are also similar in sign under factor controls. The important caveat is interpretation rather than sign stability: factor controls do not turn the high-frequency proxy into a structural shock, and the ACM decomposition remains mixed.

8.3 Narrow-Window Robustness

The narrow-window robustness check reinforces the conservative interpretation of the baseline. Coupon $[-5, +15]$ proxies preserve a 0.944 monthly correlation with the baseline z_t^{LEV}. The 10-year and 30-year same-month responses remain positive, while the 5-year descriptive reversal remains negative. Coupon $[-2, +10]$ proxies are noisier and attenuated, with monthly correlation 0.748, but they preserve the main medium- and long-maturity signs at the corresponding baseline horizons. This pattern is consistent with a market-implied supply-news proxy that contains announcement information plus ordinary intraday rate volatility. It is not consistent with treating every basis point of the event-window reaction as a clean structural supply innovation.

8.4 First-Stage Evidence

The expected-size layer is diagnostic. Let $x_t = A_t - E_{t^-} A_t$, where positive values mean larger-than-expected announced issuance, measured in billions or standardized monthly units depending on the table. Given the sign convention for z_t^{LEV}, a clean supply-quantity interpretation would predict a positive first stage. Appendix Table 20 reports first-stage relevance diagnostics across candidate endogenous issuance-surprise variables. Primary-dealer survey expectations are market-expectation inputs; TBAC recommendations are official policy benchmarks; previous-size fallbacks are diagnostic only.

Table 7 reports the LP-IV placement decision. After the expanding-standardization repair, the combined expected-size surprise does not meet the stronger evidentiary standard: the first stage is negative under the announced-minus-expected convention, the partial R^2 is below one percent, and most Anderson–Rubin confidence sets hit the grid boundary. LP-IV is therefore appendix or exploratory evidence until the sign, scaling, and weak-IV robust intervals are resolved. Survey-only surprise rows are economically useful but too small-sample for the main text. TBAC and fallback rows remain diagnostic because they are not clean pre-event market expectations.

Table 7: LP-IV readiness decision for the combined expected-size surprise. LP-IV is secondary to the direct-LP baseline and requires Anderson-Rubin intervals.

Controls	N	First-stage F	Partial R2	Sign diagnostic	AR boundary share	Decision	Paper treatment
baseline	249	2.3794	0.0107	sign_conflict_negative	0.9545	exploratory_due_to_first_stage_sign	appendix_or_deferred
factor3	249	1.6300	0.0077	sign_conflict_negative	0.9545	exploratory_due_to_first_stage_sign	appendix_or_deferred

8.5 Reading the Tables

The claims are layered. The strongest statement supported by the current direct LPs is that exact-minute Treasury supply-news announcements are associated with a positive same-month Treasury-market response, followed by negative future monthly yield-change coefficients for some maturities. Because matched non-event windows remain sizable, the baseline is described as a noisy but source-audited market-implied Treasury supply-news proxy. The credit and real-activity responses are economically plausible but less precise. Expected-size and realized-issuance tests remain diagnostic and reveal an unresolved sign/scaling tension. LP-IV estimates remain secondary or exploratory unless sign coherence and weak-IV robust confidence sets improve.

8.6 Mechanism Extensions

Table 8 summarizes the curated state-dependent mechanism LPs. This is not a second baseline model with the same sensitivity menu as Model 1. The shock proxy, lag length, and W_{t-1} controls are inherited from the direct LP; Model 2 asks whether the direct-LP response is stronger in a predetermined state. Its robustness screen is therefore narrower: states must be lagged and expanding-standardized, the interaction must have enough observations, and the sign must be stable when factor controls are added. The full interaction screen produces many candidate rows, so the main text reports only the mechanism that is both economically coherent and stable after the expanding-state repair: high rate uncertainty. In high rate-uncertainty states, the term-premium and BAA–AAA spread interactions are positive and stable under factor controls, matching preferred-habitat and uncertainty-channel interpretations. Dealer-balance-sheet and CFTC positioning interactions remain in the appendix as diagnostics because their signs or economic mapping are less robust under the current screen.

Table 8: Curated state-dependent mechanism LP summary. Rows enter the main text only when they pass the sample, sign-stability, and interpretation screens.

Mechanism	Mechanism role	Outcome	Peak horizon	Baseline interaction	Baseline HC1 EHW SE	Baseline t-stat	Baseline N	Factor3 interaction	Stable sign	Main-text worthy	Paper treatment
rate_uncertainty	Rate uncertainty amplifies the term-premium channel.	10y term-premium change	9	0.0339	0.0115	2.9452	244	0.0328	True	True	main_text
rate_uncertainty	Rate uncertainty amplifies credit-risk repricing.	BAA-AAA spread	12	0.0420	0.0162	2.5957	241	0.0562	True	True	main_text

9 Conclusion

The project is designed to be novel without losing identification discipline. The novelty is not simply adding more econometric machinery; it is assigning each method a clear role. High-frequency Treasury announcements provide a source-audited market-implied supply-news proxy. Lag-augmented LPs provide dynamic inference. Expected-size and realized-issuance measures interpret the market reaction and discipline any LP-IV scaling. Databento futures, severity screens, placebo windows, matched non-events, a volatility-regime coverage audit, and narrow-window robustness test whether the proxy is stable across market-reaction definitions.

The current evidence supports a conservative paper claim. Treasury announcement windows contain event-specific information: true release-group reactions are larger than matched non-event reactions, coupon $[-5, +15]$ proxies are highly correlated with the baseline proxy, and the medium- and long-yield LP responses keep their signs under narrower windows. At the same time, calendar-matched non-event reactions are 83.2 percent as large as true release-group reactions on average, so the signal-to-noise ratio is only about 1.20. The volatility-regime matched extension remains appendix-gated in the finalized sample because the reusable pseudo-event windows have insufficient intrawindow trade coverage. The paper therefore estimates responses to a noisy but disciplined market-implied Treasury supply-news proxy, not a fully isolated structural innovation.

The remaining extensions are used selectively. State-dependent LPs enter the current paper only where they sharpen the economic argument, currently through rate uncertainty. Dealer-balance-sheet and positioning rows remain appendix diagnostics because their economic mapping is less stable under the current screen. State-space/MCMC measurement, spectral transmission, exact CTD mapping, event-window option-IV, exact daily ETF holdings, and structured vendor expectations remain appendix-gated extensions until their diagnostics are complete. The core paper remains anchored in the audited event database, WRDS ETF proxy, falsification tests, and direct LP evidence.

A Supplementary Tables and Robustness Exhibits

This appendix records the full table and figure trail behind the main text. The main text is intentionally limited to the baseline direct-LP evidence, the matched non-event caveat, the primary narrow-window robustness check, the LP-IV placement decision, and a curated mechanism table. The supplementary exhibits below preserve auditability without turning every robustness output into a headline result. Databento, full first-stage diagnostics, full mechanism screens, lower-frequency duration precision layers, and optional course-method extensions are therefore interpreted as robustness or diagnostic evidence unless the main text explicitly says otherwise.

A.1 Event Coverage and Shock Construction

Table 9: Event and ticker coverage diagnostics for the WRDS ETF market-reaction panel.

metric	value	note
event_rows	615	primary WRDS event-factor rows
events_with_level_proxy	608	events usable for baseline monthly level proxy
severity_0_events	442	cleanest confound-screen subset
severity_0_or_1_events	615	baseline inclusion screen
ticker_reaction_rows	3868	event-ticker rows in primary WRDS reaction matrix
finite_ticker_reactions	3637	event-ticker rows with finite log returns

Table 10: Summary statistics for constructed Treasury supply-news proxies.

series	nonmissing_months	mean	std	min	max	first_month	last_month
z_level	259	-0.0677	1.1263	-5.3212	3.6803	2004-09-01	2026-04-01
z_qra_level	58	-0.2033	1.2317	-3.2297	3.0187	2010-02-01	2026-02-01
z_coupon_level	259	-0.0214	1.1502	-6.4832	4.7854	2004-09-01	2026-04-01
z_slope	258	-0.0041	1.2973	-4.5879	4.5686	2004-09-01	2026-04-01

Table 11: Baseline proxy predictability and serial-dependence diagnostics.

shock	nobs	r2	joint_wald	joint_wald_pvalue	q_stat	pvalue
z_level	259	0.0469	15.2839	0.6424	1.1740	0.9781

A.2 Deferred Appendix Extensions

State-space/MCMC measurement, spectral transmission, event-window option-IV, exact daily ETF holdings, CTD mapping, and structured vendor expectations are not standalone empirical sections in the current paper because they either lack the required diagnostics or do not change the core conclusion. The replication archive documents those deferred items, but the submitted appendix does not treat them as results.

Table 12: Placebo pre-window falsification tests for the Treasury supply-news proxy.

test	window	status	event_count	matched_true_event_count	mean_abs_placebo_level	mean_abs_true_level	placebo_to_true_abs_ratio	mean_abs_placebo_slope	corr_placebo_true_level	decision_flag
placebo_pre_window	pre_-30_-11	computed	606	606	8.3602	10.4489	0.8001	12.2761	0.2343	attention
placebo_pre_window	pre_-25_-6	computed	606	606	7.5681	10.4489	0.7243	11.2813	0.1021	pass

Table 13: Volatility-regime coverage audit. Intended matching uses lagged VIX/rate-volatility regimes with expanding-window tercile cutoffs and no full-sample regime information; skipped status means the finalized pseudo-event windows lack usable intrawindow coverage.

Status	Reason	Primary summary rows	N	Ticker rows
skipped	insufficient_intrawindow_trade_coverage_all_primary_windows_have_one_minute	230	0	0

A.3 Direct-LP Sensitivity

A.4 Expected-Size and LP-IV Diagnostics

A.5 Databento Futures Robustness

A.6 Mechanism Diagnostics

Table 14: QRA-only and coupon-only proxy diagnostics. QRA motivates the design; coupon announcements provide the broader, higher-frequency sample.

shock	nobs	mean	std	corr_with_z_level	predictability_r2	predictability_joint_wald_pvalue
z_level	259	-0.0677	1.1263	1.0000	0.0469	0.6424
z_qra_level	58	-0.2033	1.2317	0.8654	0.2349	0.9998
z_coupon_level	259	-0.0214	1.1502	0.8272	0.0644	0.3048

Table 15: Severity-zero and wide-window proxy robustness diagnostics.

baseline_shock	alternative_shock	comparison	nobs	correlation	mean_abs_difference	baseline_mean_abs	alternative_mean_abs
z_level	z_level_confound0	severity_or_window	249	0.8110	0.4410	0.8415	0.8019
z_slope	z_slope_30_60	severity_or_window	258	0.7147	0.7553	0.9842	1.0501

Table 16: Descriptive peak direct local-projection responses over horizons 0–24. Horizons are post-selected, so intervals are pointwise and not valid peak-adjusted confidence intervals.

Outcome	Peak horizon	Peak response	HC1 EHW SE	90% CI lower	90% CI upper	N
10y ACM risk-neutral yield change	1	0.0168	0.0096	0.0010	0.0326	249
10y ACM term-premium change	14	-0.0306	0.0149	-0.0552	-0.0060	239
BAA-AAA spread	2	-0.0372	0.0172	-0.0655	-0.0089	248
10y Treasury yield	0	0.0398	0.0150	0.0151	0.0645	250
2y Treasury yield	0	0.0317	0.0136	0.0092	0.0542	250
30y Treasury yield	0	0.0393	0.0128	0.0181	0.0605	250
5y Treasury yield	5	-0.0444	0.0156	-0.0702	-0.0185	247
Industrial production growth	20	-1.9732	1.5515	-4.5368	0.5905	232
Payroll growth	20	-1.6634	1.5767	-4.2686	0.9418	233
PCE inflation	1	0.3729	0.1646	0.1009	0.6448	248
Unemployment rate change	20	0.0970	0.0914	-0.0541	0.2480	231

Table 17: Direct-LP robustness to fixed-loading factor controls.

Outcome	Baseline peak horizon	Baseline peak response	Baseline 90% CI lower	Baseline 90% CI upper	Baseline N	Factor3 response at baseline peak	Factor3 90% CI lower	Factor3 90% CI upper	Factor3 N	Factor3 minus baseline
10y ACM risk-neutral yield change	1	0.0168	0.0010	0.0326	249	0.0170	0.0008	0.0332	249	0.0002
10y ACM term-premium change	14	-0.0306	-0.0552	-0.0060	239	-0.0312	-0.0550	-0.0075	239	-0.0006
10y Treasury yield	0	0.0398	0.0151	0.0645	250	0.0408	0.0142	0.0675	250	0.0010
2y Treasury yield	0	0.0317	0.0092	0.0542	250	0.0300	0.0074	0.0525	250	-0.0018
30y Treasury yield	0	0.0393	0.0181	0.0605	250	0.0398	0.0163	0.0632	250	0.0004
5y Treasury yield	5	-0.0444	-0.0702	-0.0185	247	-0.0458	-0.0704	-0.0212	247	-0.0014
BAA-AAA spread	2	-0.0372	-0.0655	-0.0089	248	-0.0452	-0.0756	-0.0148	248	-0.0080
Industrial production growth	20	-1.9732	-4.5368	0.5905	232	-2.1197	-4.8280	0.5886	232	-0.1465
PCE inflation	1	0.3729	0.1009	0.6448	248	0.3830	0.0999	0.6661	248	0.0102
Payroll growth	20	-1.6634	-4.2686	0.9418	233	-1.7690	-4.5923	1.0542	233	-0.1056
Unemployment rate change	20	0.0970	-0.0541	0.2480	231	0.1031	-0.0617	0.2678	231	0.0061

Table 18: Direct-LP lag, factor-control, and severity sensitivity at baseline peak horizons.

outcome	baseline_peak_horizon	baseline_lag3_response	baseline_lag3_nobs	baseline_lag1_response	baseline_lag2_response	baseline_lag6_response	baseline_lag12_response	factor3_lag3_response	factor5_lag3_response	severity0_lag3_response	min_sensitivity_response	max_sensitivity_response
acmrny10_diff	1	0.0168	249.0000	0.0128	0.0147	0.0102	-0.0073	0.0170	0.0175	0.0158	-0.0073	0.0175
acmtp10_diff	14	-0.0306	239.0000	-0.0243	-0.0246	-0.0368	-0.0432	-0.0312	-0.0330	-0.0513	-0.0513	-0.0243
baa_aaa_spread	2	-0.0372	248.0000	-0.0299	-0.0256	-0.0352	-0.0162	-0.0452	-0.0466	-0.0426	-0.0466	-0.0162
dgs10_diff	0	0.0398	250.0000	0.0385	0.0407	0.0380	0.0406	0.0408	0.0436	0.0512	0.0380	0.0512
dgs2_diff	0	0.0317	250.0000	0.0287	0.0299	0.0292	0.0381	0.0300	0.0345	0.0406	0.0287	0.0406
dgs30_diff	0	0.0393	250.0000	0.0389	0.0420	0.0425	0.0345	0.0398	0.0417	0.0445	0.0345	0.0445
dgs5_diff	5	-0.0444	247.0000	-0.0392	-0.0402	-0.0386	-0.0447	-0.0458	-0.0437	-0.0299	-0.0458	-0.0299
indpro_log_growth_ann	20	-1.9732	232.0000	-1.6939	-1.8243	-1.4532	-1.5383	-2.1197	-1.9109	-0.5184	-2.1197	-0.5184
payems_log_growth_ann	20	-1.6634	233.0000	-1.1743	-1.3721	-1.5120	-2.4236	-1.7690	-1.6627	-0.2097	-2.4236	-0.2097
pcepi_log_growth_ann	1	0.3729	248.0000	0.2449	0.2851	0.3507	0.2149	0.3830	0.3761	0.3027	0.2149	0.3830
unrate_diff	20	0.0970	231.0000	0.0766	0.0807	0.0947	0.1251	0.1031	0.0990	0.0134	0.0134	0.1251

Table 19: Direct-LP robustness to narrower Treasury announcement windows at baseline peak horizons.

outcome	baseline_peak_horizon	baseline_coef	baseline_nobs	narrow_shock	narrow_coef_at_baseline_peak	narrow_ci90_lower	narrow_ci90_upper	narrow_nobs	same_sign_as_baseline	abs_difference
acmrny10_diff	1	0.0168	249	z_level_coupon_5_15	0.0151	-0.0005	0.0307	249	True	0.0018
acmrny10_diff	1	0.0168	249	z_level_coupon_2_10	0.0116	-0.0028	0.0259	249	True	0.0052
acmrny10_diff	1	0.0168	249	z_level_all_5_15	0.0066	-0.0089	0.0220	249	True	0.0103
acmrny10_diff	1	0.0168	249	z_level_qra_5_15_coupon_2_10	0.0024	-0.0113	0.0162	249	True	0.0144
acmtp10_diff	14	-0.0306	239	z_level_coupon_5_15	-0.0281	-0.0519	-0.0043	239	True	0.0025
acmtp10_diff	14	-0.0306	239	z_level_coupon_2_10	-0.0125	-0.0370	0.0119	239	True	0.0181
acmtp10_diff	14	-0.0306	239	z_level_all_5_15	-0.0271	-0.0513	-0.0028	239	True	0.0035
acmtp10_diff	14	-0.0306	239	z_level_qra_5_15_coupon_2_10	-0.0102	-0.0346	0.0142	239	True	0.0204
baa_aaa_spread	2	-0.0372	248	z_level_coupon_5_15	-0.0281	-0.0557	-0.0004	248	True	0.0091
baa_aaa_spread	2	-0.0372	248	z_level_coupon_2_10	-0.0099	-0.0326	0.0129	248	True	0.0273
baa_aaa_spread	2	-0.0372	248	z_level_all_5_15	-0.0256	-0.0532	0.0020	248	True	0.0116
baa_aaa_spread	2	-0.0372	248	z_level_qra_5_15_coupon_2_10	-0.0076	-0.0305	0.0153	248	True	0.0295
dgs10_diff	0	0.0398	250	z_level_coupon_5_15	0.0353	0.0119	0.0588	250	True	0.0045
dgs10_diff	0	0.0398	250	z_level_coupon_2_10	0.0202	-0.0016	0.0420	250	True	0.0196
dgs10_diff	0	0.0398	250	z_level_all_5_15	0.0288	0.0043	0.0533	250	True	0.0110
dgs10_diff	0	0.0398	250	z_level_qra_5_15_coupon_2_10	0.0114	-0.0118	0.0346	250	True	0.0284
dgs2_diff	0	0.0317	250	z_level_coupon_5_15	0.0279	0.0074	0.0484	250	True	0.0038
dgs2_diff	0	0.0317	250	z_level_coupon_2_10	0.0177	-0.0010	0.0365	250	True	0.0140
dgs2_diff	0	0.0317	250	z_level_all_5_15	0.0225	0.0020	0.0430	250	True	0.0092
dgs2_diff	0	0.0317	250	z_level_qra_5_15_coupon_2_10	0.0113	-0.0069	0.0295	250	True	0.0204
dgs30_diff	0	0.0393	250	z_level_coupon_5_15	0.0352	0.0151	0.0553	250	True	0.0041
dgs30_diff	0	0.0393	250	z_level_coupon_2_10	0.0219	0.0025	0.0413	250	True	0.0174
dgs30_diff	0	0.0393	250	z_level_all_5_15	0.0288	0.0074	0.0503	250	True	0.0105
dgs30_diff	0	0.0393	250	z_level_qra_5_15_coupon_2_10	0.0134	-0.0078	0.0346	250	True	0.0259
dgs5_diff	5	-0.0444	247	z_level_coupon_5_15	-0.0373	-0.0656	-0.0090	247	True	0.0071
dgs5_diff	5	-0.0444	247	z_level_coupon_2_10	-0.0198	-0.0522	0.0125	247	True	0.0245
dgs5_diff	5	-0.0444	247	z_level_all_5_15	-0.0330	-0.0607	-0.0053	247	True	0.0113
dgs5_diff	5	-0.0444	247	z_level_qra_5_15_coupon_2_10	-0.0127	-0.0443	0.0189	247	True	0.0317
indpro_log_growth_ann	20	-1.9732	232	z_level_coupon_5_15	-2.2146	-5.1808	0.7516	232	True	0.2414
indpro_log_growth_ann	20	-1.9732	232	z_level_coupon_2_10	-1.9492	-4.7694	0.8710	232	True	0.0240
indpro_log_growth_ann	20	-1.9732	232	z_level_all_5_15	-2.0475	-4.6244	0.5295	232	True	0.0743
indpro_log_growth_ann	20	-1.9732	232	z_level_qra_5_15_coupon_2_10	-1.6756	-4.0682	0.7169	232	True	0.2975
payems_log_growth_ann	20	-1.6634	233	z_level_coupon_5_15	-1.8118	-4.6965	1.0730	233	True	0.1483
payems_log_growth_ann	20	-1.6634	233	z_level_coupon_2_10	-1.7465	-4.4748	0.9818	233	True	0.0831
payems_log_growth_ann	20	-1.6634	233	z_level_all_5_15	-1.5411	-4.0001	0.9180	233	True	0.1223
payems_log_growth_ann	20	-1.6634	233	z_level_qra_5_15_coupon_2_10	-1.4313	-3.6670	0.8045	233	True	0.2322
pcepi_log_growth_ann	1	0.3729	248	z_level_coupon_5_15	0.3474	0.0816	0.6132	248	True	0.0255
pcepi_log_growth_ann	1	0.3729	248	z_level_coupon_2_10	0.2655	0.0274	0.5037	248	True	0.1073
pcepi_log_growth_ann	1	0.3729	248	z_level_all_5_15	0.2666	0.0010	0.5322	248	True	0.1063
pcepi_log_growth_ann	1	0.3729	248	z_level_qra_5_15_coupon_2_10	0.1756	-0.0624	0.4137	248	True	0.1972
unrate_diff	20	0.0970	231	z_level_coupon_5_15	0.1132	-0.0596	0.2860	231	True	0.0162
unrate_diff	20	0.0970	231	z_level_coupon_2_10	0.1103	-0.0530	0.2736	231	True	0.0133
unrate_diff	20	0.0970	231	z_level_all_5_15	0.0989	-0.0474	0.2451	231	True	0.0019
unrate_diff	20	0.0970	231	z_level_qra_5_15_coupon_2_10	0.0927	-0.0404	0.2258	231	True	0.0043

Table 20: Expected-size first-stage relevance diagnostics. Survey rows are market-expectation inputs; TBAC and fallback rows are diagnostic.

endogenous_variable	instrument	control_set	lag_count	nobs	coef_on_instrument	se_ehw	t_stat	robust_wald_stat	robust_wald_pvalue	classical_first_stage_f	partial_r2	sample_start	sample_end	expected_sign	sign_diagnostic	robust_relevance_flag	diagnostic_label	strong_first_stage	small_sample_promising
monthly_supply_surprise_billions	z_level	baseline	3	250	-17.5663	13.8254	-1.2706	1.6144	0.2039	5.9430	0.0258	2004-12-01	2026-04-01	positive	sign_conflict_negative	0	sign_conflict_exploratory	0	0
monthly_supply_surprise_qra_billions	z_level	baseline	3	12	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2018-05-01	2024-05-01	positive	not_assessed	0	weak_or_insufficient	0	0
monthly_supply_surprise_coupon_billions	z_level	baseline	3	250	2.4015	1.1615	2.0676	4.2749	0.0387	4.0518	0.0178	2004-12-01	2026-04-01	positive	coherent_positive	1	weak_or_insufficient	0	0
monthly_supply_surprise_survey_billions	z_level	baseline	3	38	-256.2426	57.3127	-4.4710	19.9894	0.0000	20.3992	0.6296	2018-05-01	2024-07-01	positive	sign_conflict_negative	1	sign_conflict_exploratory	0	0
monthly_supply_surprise_fallback_billions	z_level	baseline	3	243	0.7750	0.9512	0.8147	0.6638	0.4152	0.6218	0.0029	2004-12-01	2026-04-01	diagnostic	diagnostic_no_expected_sign	0	weak_or_insufficient	0	0
monthly_tbac_surprise_billions	z_level	baseline	3	102	-0.3946	0.1504	-2.6235	6.8828	0.0087	6.2230	0.0757	2004-12-01	2019-09-01	diagnostic	diagnostic_no_expected_sign	1	weak_or_insufficient	0	0
monthly_tbac_surprise_coupon_billions	z_level	baseline	3	102	-0.3946	0.1504	-2.6235	6.8828	0.0087	6.2230	0.0757	2004-12-01	2019-09-01	diagnostic	diagnostic_no_expected_sign	1	weak_or_insufficient	0	0
monthly_duration_equiv_supply_surprise	z_level	baseline	3	250	4.8923	5.8175	0.8410	0.7072	0.4004	0.7858	0.0035	2004-12-01	2026-04-01	positive	coherent_positive	0	weak_or_insufficient	0	0
monthly_duration_equiv_tbac_surprise	z_level	baseline	3	102	-1.3149	0.5027	-2.6158	6.8424	0.0089	6.1531	0.0749	2004-12-01	2019-09-01	diagnostic	diagnostic_no_expected_sign	1	weak_or_insufficient	0	0

Table 21: Databento futures symbol screen for appendix robustness specifications.

ticker	symbol_group	overlap_event_count	nonmissing_event_count	nonmissing_event_share	median_coverage_ratio	zero_volume_or_zero_return_share	first_usable_event_date	screen_label	passes_nonmissing	passes_coverage	passes_zero_activity	passes_start_date
ZT.v.0	core_treasury_future	1102	1102	1.0000	1.0000	0.1960	2011-01-06	paper_usable	1	1	1	1
ZF.v.0	core_treasury_future	1102	1102	1.0000	1.0000	0.0662	2011-01-06	paper_usable	1	1	1	1
ZN.v.0	core_treasury_future	1102	1102	1.0000	1.0000	0.0935	2011-01-06	paper_usable	1	1	1	1
ZB.v.0	core_treasury_future	1102	1102	1.0000	1.0000	0.0871	2011-01-06	paper_usable	1	1	1	1
UB.v.0	core_treasury_future	1102	1102	1.0000	1.0000	0.0272	2011-01-06	paper_usable	1	1	1	1
TN.v.0	sparse_candidate	1102	811	0.7359	1.0000	0.0666	2011-01-06	appendix_only	0	1	1	1
ZQ.v.0	sparse_candidate	1102	949	0.8612	0.1951	0.6228	2011-01-06	appendix_only	1	0	0	1
SR3.v.0	sparse_candidate	1102	496	0.4501	0.8537	0.3589	2018-05-17	excluded	0	1	0	0

Table 22: WRDS ETF and Databento futures overlap correlations. Databento is robustness evidence over the 2011–2026 overlap.

sample	shock_type	nobs	correlation	futures_mean	wrds_mean	futures_sd	wrds_sd
all	level	1102	0.7237	0.0344	-0.3759	10.4619	15.2782
all	slope	1063	0.7441	-0.0940	-0.3300	15.9781	23.8908
qra	level	49	0.8029	1.7543	0.6511	14.1051	25.7661
qra	slope	40	0.8428	1.8569	2.5156	26.4638	44.4750
coupon	level	1053	0.7172	-0.0456	-0.4237	10.2629	14.6271
coupon	slope	1023	0.7321	-0.1702	-0.4412	15.4406	22.7442

Table 23: Largest WRDS-Databento disagreement events, used to diagnose market-proxy fragility.

event_id	event_date_databento	security_type_databento	security_term_databento	futures_level_proxy	level_proxy	level_difference_futures_minus_wrds	standardized_level_disagreement	abs_standardized_level_disagreement	futures_slope_proxy	slope_proxy	slope_difference_futures_minus_wrds	standardized_slope_disagreement	cluster_covid_window	cluster_qra	cluster_recent_2026	futures_level_tickers_used	futures_coverage_ratio_mean
auction_announcement_20200319_note_1_year_10_month_912828z45	2020-03-19	Note	1-Year 10-Month	31.4219	-25.8025	57.2244	5.4505	5.4505	51.0026	-47.9168	98.9194	6.4979	True	False	False	TN.v.0,UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.7988
auction_announcement_20200319_note_2_year_912828zg8	2020-03-19	Note	2-Year	31.4219	-25.8025	57.2244	5.4048	5.4048	51.0026	-47.9168	98.9194	6.4279	True	False	False	TN.v.0,UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.7988
auction_announcement_20200319_note_5_year_912828zf0	2020-03-19	Note	5-Year	31.4219	-25.8025	57.2244	5.3605	5.3605	51.0026	-47.9168	98.9194	6.3604	True	False	False	TN.v.0,UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.7988
auction_announcement_20200319_note_7_year_912828ze3	2020-03-19	Note	7-Year	31.4219	-25.8025	57.2244	5.3174	5.3174	51.0026	-47.9168	98.9194	6.2952	True	False	False	TN.v.0,UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.7988
auction_announcement_20250205_bond_30_year_912810ug1	2025-02-05	Bond	30-Year	7.9761	-32.8536	40.8296	2.8936	2.8936	-0.5704	-15.9417	15.3713	0.6173	False	False	False	TN.v.0,UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.8963
auction_announcement_20250205_note_10_year_91282cmm0	2025-02-05	Note	10-Year	7.9761	-32.8536	40.8296	2.8873	2.8873	-0.5704	-15.9417	15.3713	0.6168	False	False	False	TN.v.0,UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.8963
auction_announcement_20250205_note_3_year_91282cmn8	2025-02-05	Note	3-Year	7.9761	-32.8536	40.8296	2.8810	2.8810	-0.5704	-15.9417	15.3713	0.6164	False	False	False	TN.v.0,UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.8963
qra_archive_reference_2025q1_0006	2025-02-05	NaN	NaN	7.9761	-32.8536	40.8296	2.8748	2.8748	-0.5704	-15.9417	15.3713	0.6159	False	False	False	TN.v.0,UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.8963
auction_announcement_20200903_note_3_year_91282cak7	2020-09-03	Note	3-Year	0.5956	-35.2048	35.8003	2.5092	2.5092	0.1941	-56.8654	57.0595	2.6993	False	False	False	TN.v.0,UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.7774
auction_announcement_20231101_bond_30_year_912810tv0	2023-11-01	Bond	30-Year	-21.5974	-66.8164	45.2190	2.3785	2.3785	-52.9989	-144.6877	91.6888	3.0668	False	False	False	TN.v.0,UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.9085
auction_announcement_20231101_note_10_year_91282cjj1	2023-11-01	Note	10-Year	-21.5974	-66.8164	45.2190	2.3342	2.3342	-52.9989	-144.6877	91.6888	2.9386	False	False	False	TN.v.0,UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.9085
auction_announcement_20120105_note_3_year_912828rz5	2012-01-05	Note	3-Year	22.8865	-1.4653	24.3519	2.3044	2.3044	34.7201	-4.9666	39.6867	2.6326	False	False	False	UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.8293
auction_announcement_20231101_note_3_year_91282cjk8	2023-11-01	Note	3-Year	-21.5974	-66.8164	45.2190	2.2915	2.2915	-52.9989	-144.6877	91.6888	2.8201	False	False	False	TN.v.0,UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.9085
auction_announcement_20230201_bond_30_year_912810tn8	2023-02-01	Bond	30-Year	-11.5211	16.3641	-27.8852	-2.2714	2.2714	-16.7775	44.0796	-60.8571	-3.0985	False	False	False	TN.v.0,UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.9207
auction_announcement_20230201_note_10_year_91282cgm7	2023-02-01	Note	10-Year	-11.5211	16.3641	-27.8852	-2.2684	2.2684	-16.7775	44.0796	-60.8571	-3.0909	False	False	False	TN.v.0,UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.9207
auction_announcement_20230201_note_3_year_91282cgl9	2023-02-01	Note	3-Year	-11.5211	16.3641	-27.8852	-2.2654	2.2654	-16.7775	44.0796	-60.8571	-3.0834	False	False	False	TN.v.0,UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.9207
qra_archive_reference_2023q1_0014	2023-02-01	NaN	NaN	-11.5211	16.3641	-27.8852	-2.2624	2.2624	-16.7775	44.0796	-60.8571	-3.0760	False	False	False	TN.v.0,UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.9207
qra_archive_reference_2023q4_0011	2023-11-01	NaN	NaN	-21.5974	-66.8164	45.2190	2.2504	2.2504	-52.9989	-144.6877	91.6888	2.7101	False	False	False	TN.v.0,UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.9085
auction_announcement_20160203_bond_30_year_912810rq3	2016-02-03	Bond	30-Year	2.0252	32.2695	-30.2443	-2.1273	2.1273	7.9603	64.5389	-56.5786	-2.6442	False	False	False	TN.v.0,UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.9129
auction_announcement_20160203_note_10_year_912828p46	2016-02-03	Note	10-Year	2.0252	32.2695	-30.2443	-2.1072	2.1072	7.9603	64.5389	-56.5786	-2.5857	False	False	False	TN.v.0,UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.9129
auction_announcement_20160203_note_3_year_912828p53	2016-02-03	Note	3-Year	2.0252	32.2695	-30.2443	-2.0876	2.0876	7.9603	64.5389	-56.5786	-2.5303	False	False	False	TN.v.0,UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.9129
qra_archive_reference_2016q1_0042	2016-02-03	NaN	NaN	2.0252	32.2695	-30.2443	-2.0686	2.0686	7.9603	64.5389	-56.5786	-2.4780	False	False	False	TN.v.0,UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.9129
auction_announcement_20140717_note_10_year_912828wu0	2014-07-17	Note	10-Year	-23.4489	-13.6913	-9.7575	-2.0089	2.0089	-25.6324	-23.2444	-2.3881	-1.2574	False	False	False	UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.8008
auction_announcement_20200402_note_3_year_912828zh6	2020-04-02	Note	3-Year	-2.8019	21.7888	-24.5906	-1.9970	1.9970	-3.4225	31.5079	-34.9304	-1.9023	True	False	False	TN.v.0,UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.7683
auction_announcement_20250522_note_1_year_11_month_91282cmx6	2025-05-22	Note	1-Year 11-Month	-17.9211	3.6211	-21.5422	-1.9941	1.9941	-28.4395	3.8011	-32.2406	-1.9675	False	False	False	TN.v.0,UB.v.0,ZB.v.0,ZF.v.0,ZN.v.0	0.8811

Table 24: Databento futures-proxy direct-LP robustness summary. These are not CTD-adjusted yield-equivalent estimates.

Outcome	Futures shock	Peak horizon	Futures response	Futures 90% CI lower	Futures 90% CI upper	Futures N	WRDS response at same horizon	Same sign as WRDS	Paper treatment
10y risk-neutral change	z_futures_level_screened	24	0.0199	-0.0004	0.0401	145	0.0060	True	robustness_only
10y term-premium change	z_futures_level_screened	23	-0.0227	-0.0411	-0.0043	146	-0.0188	True	robustness_only
BAA-AAA	z_futures_level_screened	1	0.0215	-0.0061	0.0491	165	-0.0286	False	robustness_only
10y yield	z_futures_level_screened	5	-0.0386	-0.0625	-0.0147	163	-0.0367	True	robustness_only
2y yield	z_futures_level_screened	4	-0.0346	-0.0572	-0.0121	163	-0.0235	True	robustness_only
30y yield	z_futures_level_screened	5	-0.0312	-0.0538	-0.0085	163	-0.0291	True	robustness_only
5y yield	z_futures_level_screened	5	-0.0447	-0.0697	-0.0197	163	-0.0444	True	robustness_only
Industrial production	z_futures_level_screened	20	-2.6708	-6.3490	1.0074	148	-1.9732	True	robustness_only
Payrolls	z_futures_level_screened	1	-3.1254	-8.6408	2.3900	165	0.6222	False	robustness_only
PCE inflation	z_futures_level_screened	24	0.3364	0.0870	0.5858	144	0.0070	True	robustness_only
Unemployment	z_futures_level_screened	1	0.1836	-0.1423	0.5095	162	-0.0301	False	robustness_only
10y risk-neutral change	z_futures_slope_screened	5	-0.0203	-0.0320	-0.0086	163	-0.0159	True	robustness_only
10y term-premium change	z_futures_slope_screened	0	0.0237	0.0017	0.0457	166	0.0130	True	robustness_only
BAA-AAA	z_futures_slope_screened	1	0.0215	-0.0049	0.0478	165	-0.0286	False	robustness_only
10y yield	z_futures_slope_screened	5	-0.0367	-0.0598	-0.0135	163	-0.0367	True	robustness_only
2y yield	z_futures_slope_screened	4	-0.0313	-0.0514	-0.0112	163	-0.0235	True	robustness_only
30y yield	z_futures_slope_screened	5	-0.0304	-0.0520	-0.0088	163	-0.0291	True	robustness_only
5y yield	z_futures_slope_screened	5	-0.0415	-0.0654	-0.0176	163	-0.0444	True	robustness_only
Industrial production	z_futures_slope_screened	20	-2.1936	-5.2782	0.8911	148	-1.9732	True	robustness_only
Payrolls	z_futures_slope_screened	1	-3.0831	-8.4748	2.3086	165	0.6222	False	robustness_only
PCE inflation	z_futures_slope_screened	24	0.3268	0.0900	0.5636	144	0.0070	True	robustness_only
Unemployment	z_futures_slope_screened	1	0.1847	-0.1372	0.5066	162	-0.0301	False	robustness_only

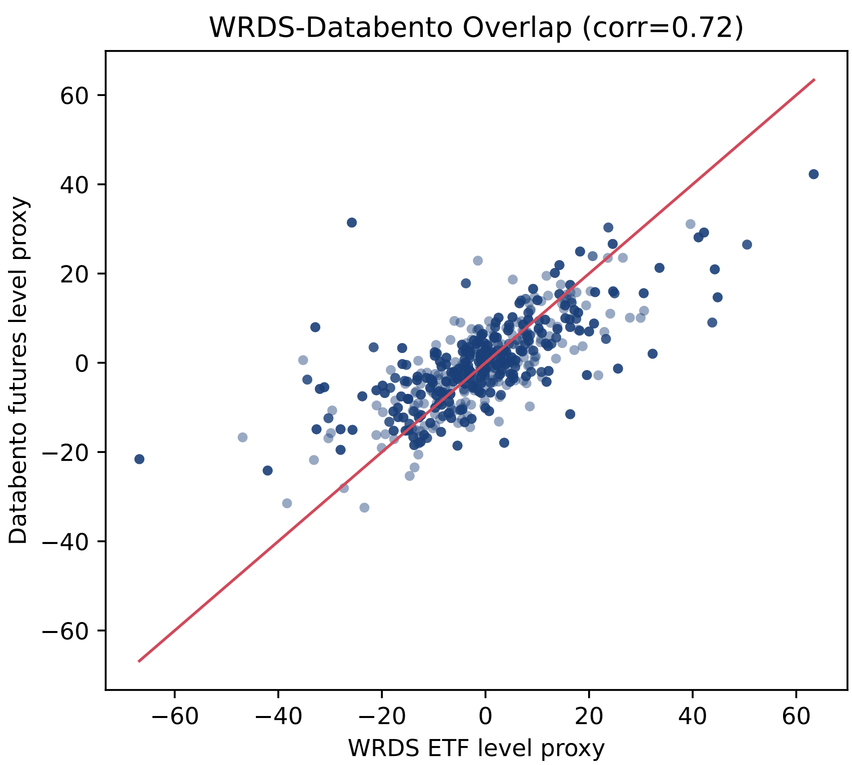

Figure 7: Event-level overlap between the WRDS ETF level proxy and the screened Databento futures level proxy. Databento is robustness evidence over 2011–2026, not a splice into the WRDS baseline.

Table 25: Full state-dependent mechanism LP decision table retained for appendix auditability.

Mechanism	State variable	Outcome	Peak horizon	Baseline interaction	Baseline HC1 EHW SE	Baseline t-stat	Baseline N	Factor3 interaction	Factor3 N	Stable sign	Main-text worthy	Paper treatment
cftc_positioning	dealer_position_state_lag1	10y risk-neutral change	3	-0.0256	0.0106	-2.4015	172	-0.0259	172	True	True	candidate_main_text
cftc_positioning	leveraged_money_duration_state_lag1	10y risk-neutral change	16	0.0164	0.0085	1.9389	162	0.0173	162	True	True	candidate_main_text
cftc_positioning	dealer_position_state_lag1	10y term-premium change	17	0.0456	0.0233	1.9556	161	0.0428	161	True	True	candidate_main_text
cftc_positioning	leveraged_money_duration_state_lag1	10y term-premium change	17	0.0374	0.0155	2.4179	161	0.0397	161	True	True	candidate_main_text
cftc_positioning	dealer_position_state_lag1	10y yield	4	-0.0557	0.0266	-2.0967	172	-0.0525	172	True	True	candidate_main_text
cftc_positioning	leveraged_money_duration_state_lag1	10y yield	1	0.0355	0.0177	2.0087	175	0.0277	175	True	True	candidate_main_text
cftc_positioning	dealer_position_state_lag1	2y yield	4	-0.0737	0.0215	-3.4307	172	-0.0748	172	True	True	candidate_main_text
cftc_positioning	leveraged_money_duration_state_lag1	2y yield	4	-0.0276	0.0161	-1.7153	172	-0.0263	172	True	True	candidate_main_text
cftc_positioning	leveraged_money_duration_state_lag1	30y yield	1	0.0349	0.0159	2.1973	175	0.0285	175	True	True	candidate_main_text
cftc_positioning	dealer_position_state_lag1	5y yield	4	-0.0733	0.0254	-2.8829	172	-0.0716	172	True	True	candidate_main_text
cftc_positioning	leveraged_money_duration_state_lag1	5y yield	1	0.0337	0.0184	1.8309	175	0.0264	175	True	True	candidate_main_text
cftc_positioning	dealer_position_state_lag1	BAA-AAA spread	15	0.0321	0.0156	2.0633	163	0.0344	163	True	True	candidate_main_text
cftc_positioning	leveraged_money_duration_state_lag1	BAA-AAA spread	18	0.0217	0.0125	1.7397	160	0.0239	160	True	True	candidate_main_text
cftc_positioning	dealer_position_state_lag1	Corporate return: HY	20	-0.0049	0.0023	-2.1661	145	-0.0051	145	True	True	candidate_main_text
cftc_positioning	dealer_position_state_lag1	Corporate return: IG	3	0.0048	0.0018	2.6093	162	0.0050	162	True	True	candidate_main_text
cftc_positioning	leveraged_money_duration_state_lag1	Corporate return: IG	17	-0.0034	0.0015	-2.3012	148	-0.0033	148	True	True	candidate_main_text
cftc_positioning	dealer_position_state_lag1	Corporate return: all	3	0.0046	0.0018	2.5319	162	0.0049	162	True	True	candidate_main_text
cftc_positioning	leveraged_money_duration_state_lag1	Corporate return: all	17	-0.0032	0.0015	-2.1279	148	-0.0030	148	True	True	candidate_main_text
cftc_positioning	dealer_position_state_lag1	Corporate spread: HY	24	-0.0028	0.0017	-1.6535	138	-0.0016	138	True	True	candidate_main_text
cftc_positioning	leveraged_money_duration_state_lag1	Corporate spread: IG	5	-0.0007	0.0003	-1.9997	157	-0.0007	157	True	True	candidate_main_text
cftc_positioning	dealer_position_state_lag1	Corporate spread: all	15	0.0011	0.0005	1.9408	147	0.0010	147	True	True	candidate_main_text
cftc_positioning	dealer_position_state_lag1	PCE inflation	15	-0.4440	0.2247	-1.9764	162	-0.5327	162	True	True	candidate_main_text
cftc_positioning	leveraged_money_duration_state_lag1	PCE inflation	3	-0.3174	0.1444	-2.1973	172	-0.3507	172	True	True	candidate_main_text
dealer_balance_sheet	dealer_constraint_state_lag1	10y risk-neutral change	7	0.0479	0.0190	2.5261	138	0.0403	138	True	True	candidate_main_text
dealer_balance_sheet	dealer_constraint_state_lag1	10y term-premium change	17	0.0637	0.0234	2.7218	128	0.0653	128	True	True	candidate_main_text
dealer_balance_sheet	dealer_constraint_state_lag1	10y yield	17	0.0623	0.0337	1.8498	129	0.0635	129	True	True	candidate_main_text
dealer_balance_sheet	dealer_constraint_state_lag1	2y yield	19	0.0905	0.0510	1.7736	127	0.0762	127	True	True	candidate_main_text
dealer_balance_sheet	dealer_constraint_state_lag1	BAA-AAA spread	20	-0.0558	0.0328	-1.7024	125	-0.0124	125	True	True	candidate_main_text
dealer_balance_sheet	dealer_constraint_state_lag1	Corporate return: HY	9	0.0065	0.0037	1.7338	123	0.0066	123	True	True	candidate_main_text
dealer_balance_sheet	dealer_constraint_state_lag1	Corporate return: all	15	0.0055	0.0030	1.8448	117	0.0043	117	True	True	candidate_main_text
dealer_balance_sheet	dealer_constraint_state_lag1	Corporate spread: IG	13	0.0007	0.0003	2.0832	116	0.0003	116	True	True	candidate_main_text
dealer_balance_sheet	dealer_constraint_state_lag1	Corporate spread: all	13	0.0011	0.0006	1.8083	116	0.0005	116	True	True	candidate_main_text
dealer_balance_sheet	dealer_constraint_state_lag1	PCE inflation	9	0.5632	0.3382	1.6651	135	0.4590	135	True	True	candidate_main_text
rate_uncertainty	rate_vol_state_lag1	10y risk-neutral change	6	-0.0212	0.0098	-2.1681	247	-0.0214	247	True	True	candidate_main_text
rate_uncertainty	rate_vol_state_lag1	10y term-premium change	9	0.0339	0.0115	2.9452	244	0.0328	244	True	True	candidate_main_text
rate_uncertainty	rate_vol_state_lag1	10y yield	0	0.0293	0.0124	2.3653	250	0.0247	250	True	True	candidate_main_text
rate_uncertainty	rate_vol_state_lag1	2y yield	22	0.0223	0.0100	2.2331	232	0.0241	232	True	True	candidate_main_text
rate_uncertainty	rate_vol_state_lag1	30y yield	0	0.0285	0.0108	2.6408	250	0.0253	250	True	True	candidate_main_text
rate_uncertainty	rate_vol_state_lag1	BAA-AAA spread	12	0.0420	0.0162	2.5957	241	0.0562	241	True	True	candidate_main_text
rate_uncertainty	rate_vol_state_lag1	Corporate return: HY	8	0.0032	0.0013	2.4028	232	0.0037	232	True	True	candidate_main_text
rate_uncertainty	rate_vol_state_lag1	Corporate return: IG	10	-0.0017	0.0008	-2.2281	230	-0.0019	230	True	True	candidate_main_text
rate_uncertainty	rate_vol_state_lag1	Corporate return: all	4	-0.0020	0.0008	-2.5727	236	-0.0023	236	True	True	candidate_main_text
rate_uncertainty	rate_vol_state_lag1	Corporate spread: HY	7	0.0045	0.0020	2.2997	230	0.0047	230	True	True	candidate_main_text
rate_uncertainty	rate_vol_state_lag1	Corporate spread: IG	6	0.0011	0.0004	2.5542	231	0.0012	231	True	True	candidate_main_text
rate_uncertainty	rate_vol_state_lag1	Corporate spread: all	6	0.0020	0.0008	2.6247	231	0.0021	231	True	True	candidate_main_text
rate_uncertainty	rate_vol_state_lag1	PCE inflation	6	-0.3240	0.1348	-2.4034	246	-0.3321	246	True	True	candidate_main_text
cftc_positioning	dealer_position_state_lag1	30y yield	21	0.0336	0.0236	1.4233	158	0.0295	158	True	False	appendix_or_omit
cftc_positioning	leveraged_money_duration_state_lag1	Corporate return: HY	19	0.0023	0.0029	0.8102	146	0.0017	146	True	False	appendix_or_omit
cftc_positioning	leveraged_money_duration_state_lag1	Corporate spread: HY	20	-0.0016	0.0013	-1.1854	142	-0.0013	142	True	False	appendix_or_omit
cftc_positioning	dealer_position_state_lag1	Corporate spread: IG	14	0.0005	0.0004	1.2591	148	0.0004	148	True	False	appendix_or_omit
cftc_positioning	leveraged_money_duration_state_lag1	Corporate spread: all	5	-0.0008	0.0006	-1.3707	157	-0.0008	157	True	False	appendix_or_omit
cftc_positioning	dealer_position_state_lag1	Industrial production	15	-3.8799	3.1512	-1.2312	162	-4.1400	162	True	False	appendix_or_omit
cftc_positioning	leveraged_money_duration_state_lag1	Industrial production	20	4.2176	3.2638	1.2922	157	4.4192	157	True	False	appendix_or_omit
cftc_positioning	dealer_position_state_lag1	Payrolls	15	-3.0381	3.0177	-1.0068	163	-3.4312	163	True	False	appendix_or_omit
cftc_positioning	leveraged_money_duration_state_lag1	Payrolls	20	3.4710	3.1136	1.1148	158	3.8459	158	True	False	appendix_or_omit
cftc_positioning	dealer_position_state_lag1	Unemployment	15	0.1893	0.1905	0.9938	161	0.2088	161	True	False	appendix_or_omit
cftc_positioning	leveraged_money_duration_state_lag1	Unemployment	20	-0.2267	0.1949	-1.1629	156	-0.2499	156	True	False	appendix_or_omit
dealer_balance_sheet	dealer_constraint_state_lag1	30y yield	9	0.0568	0.0365	1.5560	137	0.0494	137	True	False	appendix_or_omit
dealer_balance_sheet	dealer_constraint_state_lag1	5y yield	19	0.0785	0.0514	1.5258	127	0.0635	127	True	False	appendix_or_omit
dealer_balance_sheet	dealer_constraint_state_lag1	Corporate return: IG	10	-0.0057	0.0037	-1.5277	122	-0.0041	122	True	False	appendix_or_omit
dealer_balance_sheet	dealer_constraint_state_lag1	Corporate spread: HY	7	0.0034	0.0032	1.0652	122	0.0004	122	True	False	appendix_or_omit
dealer_balance_sheet	dealer_constraint_state_lag1	Industrial production	20	5.2033	5.2655	0.9882	124	4.8522	124	True	False	appendix_or_omit
dealer_balance_sheet	dealer_constraint_state_lag1	Payrolls	20	5.2774	5.0575	1.0435	125	5.1716	125	True	False	appendix_or_omit
dealer_balance_sheet	dealer_constraint_state_lag1	Unemployment	20	-0.3151	0.3264	-0.9653	123	-0.3145	123	True	False	appendix_or_omit
rate_uncertainty	rate_vol_state_lag1	5y yield	19	0.0250	0.0160	1.5671	235	0.0231	235	True	False	appendix_or_omit
rate_uncertainty	rate_vol_state_lag1	Industrial production	0	-1.7410	1.1710	-1.4867	249	-1.8233	249	True	False	appendix_or_omit
rate_uncertainty	rate_vol_state_lag1	Payrolls	0	-1.8565	1.1840	-1.5680	250	-1.9984	250	True	False	appendix_or_omit
rate_uncertainty	rate_vol_state_lag1	Unemployment	0	0.1114	0.0732	1.5217	248	0.1126	248	True	False	appendix_or_omit

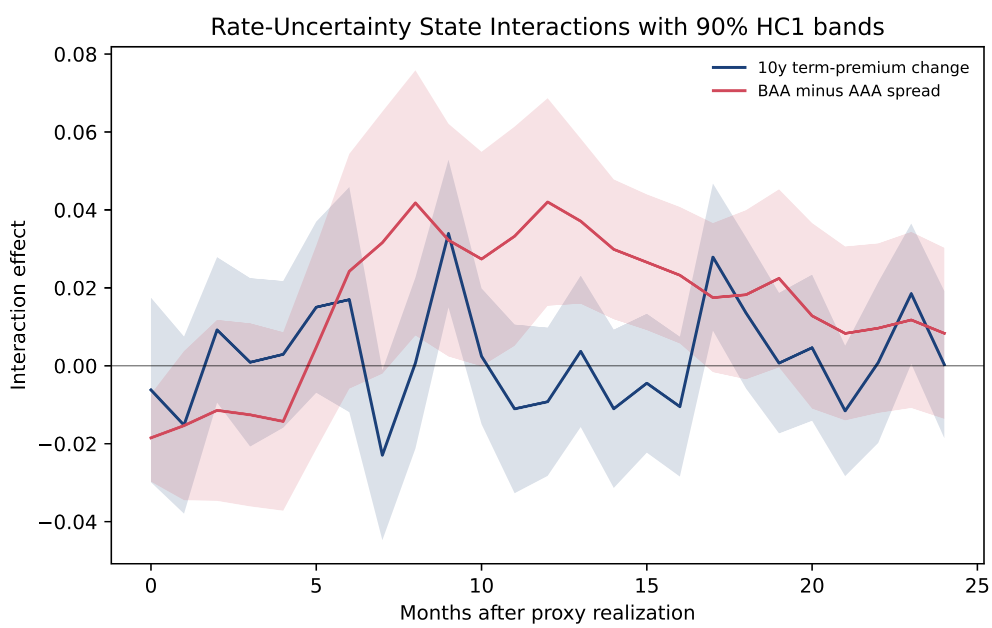

Figure 8: Rate-uncertainty state interactions. The state variable is lagged one month, so these are conditional projection effects, not a separate identification source or a structural nonlinear impulse response.

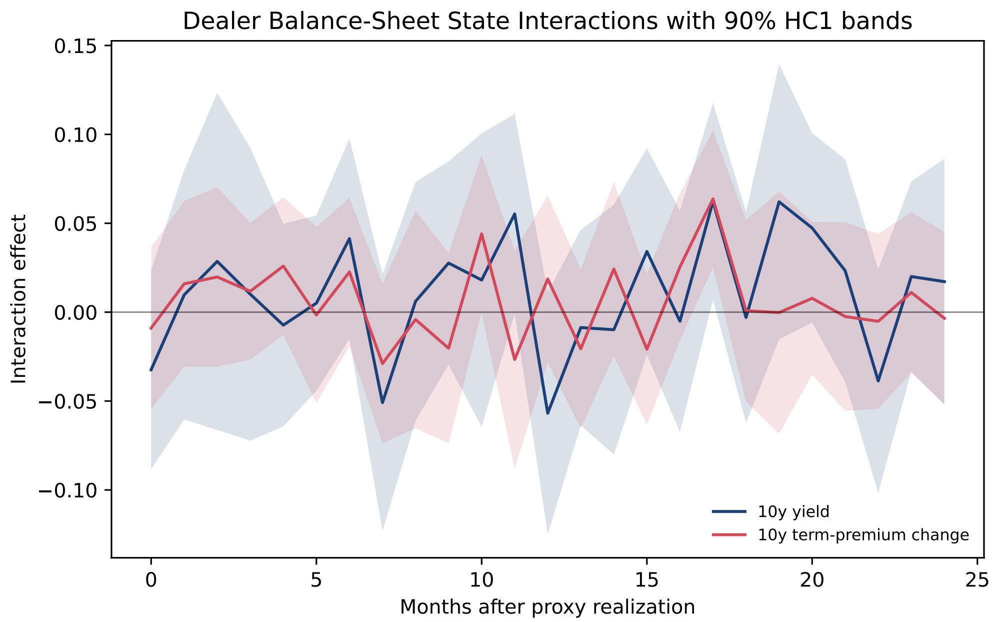

Figure 9: Dealer-balance-sheet state interactions. These rows are retained as appendix diagnostics. After the expanding-state repair, the curated main-text mechanism table keeps only rate-uncertainty interactions because the dealer rows do not supply a stable, economically clean amplification story.

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